KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN
[ACCREDITED BY NAAC $ AN ISO 9001:2015 CERTIFIED INSTITUTION]
DEPARTMENT OF COMPUTER SCIENCE
UG-COMPUTER SCIENCE [2025-2026]

COURIER MANAGEMENT SYSTEM

Project Work Done By

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DEPARTMENT OF COMPUTER SCIENCE

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

(Permanently Affiliated to Thiruvalluvar University)

(Accredited by NAAC & An ISO 9001:2015 Certified Institution)

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Cricket Player Performance Tracking Application: CRICKSTATS

User Story:

As a Salesforce Administrator, the main goal of this project is to develop a comprehensive cricket player performance tracking system within Salesforce. The application is designed to capture and manage detailed match-by-match performance statistics of cricket players.The CRICKSTATS system uses custom objects such as Player and Match Performance, which store various player-related data including runs scored, wickets taken, matches played, and other performance statistics. These objects are connected using relationships such as Master-Detail and Lookup Relationships, which ensure proper data organization and integrity.This application helps coaches, analysts, and team managers to easily monitor player performance, generate reports, and analyze performance trends. By using dashboards and reports, users can visualize player statistics and make strategic decisions related to team selection, player imp Cricket is one of the most popular sports in the world. In modern cricket, performance analysis plays a very important role in improving player skills and team strategy.

1.INTRODUCTION

1.1 Project Overview:

CRICKSTATS is a cloud-based application developed using the Salesforce platform to track and analyze cricket player performance. Cricket teams often maintain performance records manually in notebooks or spreadsheets. This process can lead to data loss, incorrect calculations, and difficulty in analyzing player performance.To solve these issues, the CRICKSTATS system provides a centralized digital platform where player statistics can be stored, updated, and analyzed efficiently.

The application uses Salesforce features such as custom objects**,** automation flows**,** reports**,** and dashboards to create an effective performance management system.The system allows users to enter match data, update player statistics automatically, and generate detailed reports that help evaluate player performance over time.

1.2 Purpose :

The main purpose of the CRICKSTATS project is to provide a digital solution for managing cricket player performance data.

The objectives of the system include:

- To convert manual player statistics into a digital system
- To improve the accuracy of cricket performance records
- To provide real-time data analysis and reporting
- To help coaches and analysts evaluate player performance
- To automate record management using Salesforce tools
- To provide a centralized database for storing player and match information

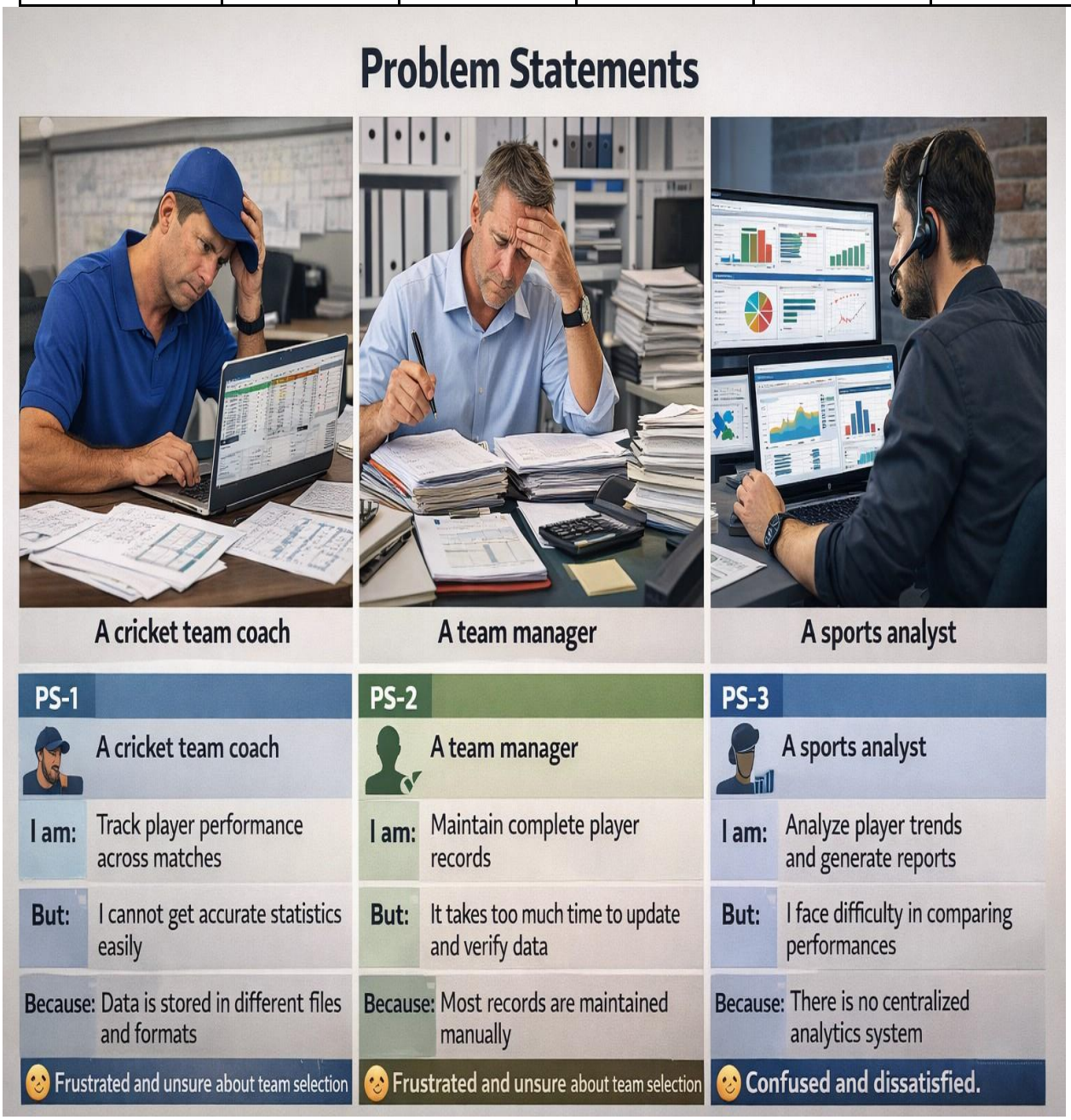
This system helps cricket teams make better decisions based on accurate performance data.

2. IDEATION PHASE

2.1 Problem Statement(Customer Problem Statement Template)

Problem Statement (PS)	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	A cricket team coach	Track player performance across matches	I cannot get accurate statistics easily	Data is stored in different files and formats	Frustrated and unsure about team selection

PS-2	A team manager	Maintain complete player records	It takes too much time to update and verify data	Most records are maintained manually	Stressed and inefficient



1.1 Empathy Map Canvas

Target Users

- Team Managers
- Coaches
- Analysts
- Salesforce Admins

Empathy Map Analysis

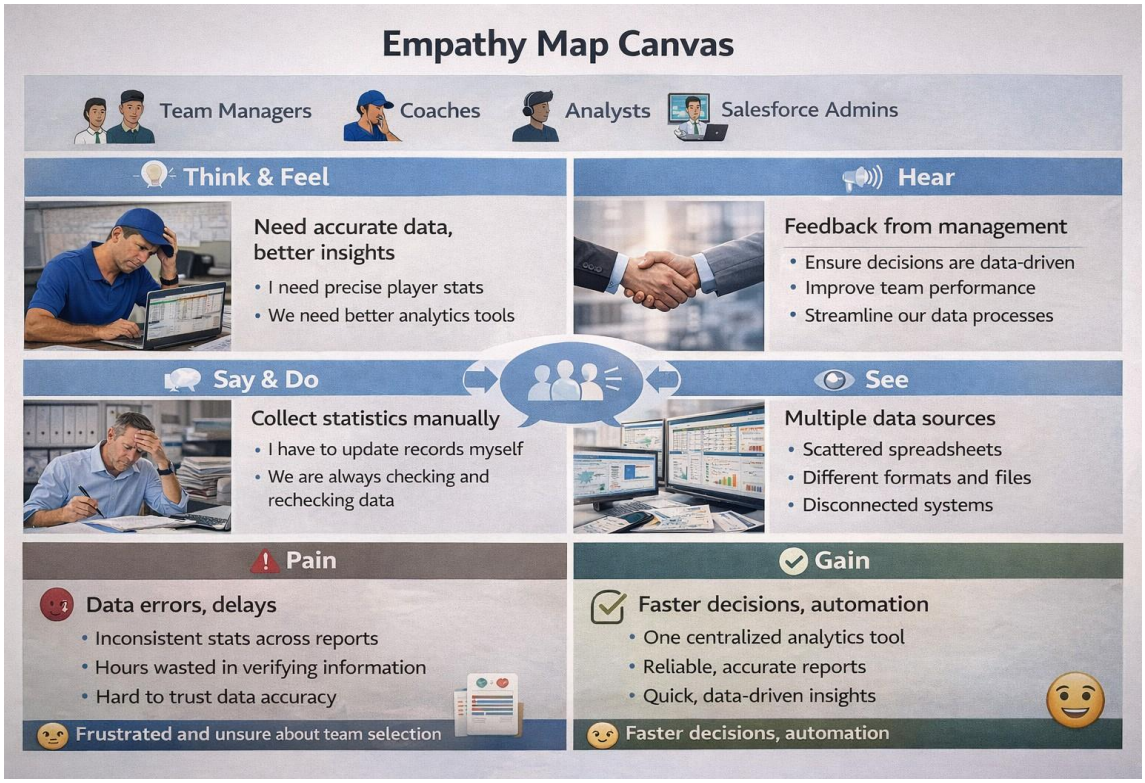
Aspect	Description

1. Think&feel : Users need accurate player data and better insights to make decisions
2. Hear: Coaches and managers hear feedback from team owners, management, and players regarding performance and match results.
3. See: Users see player statistics stored in different files, spreadsheets, and manual records which makes analysis difficult.
4. Say & Do: They usually collect match statistics manually and maintain player records using notebooks or spreadsheets.
5. Pain: Errors in data, delays in updating statistics, and difficulty in analyzing player performance.
6. Gain: Faster decision-making, automated data management, and accurate player performance analysis.

- **Empathy Analysis**

Aspect **Description**

- **Think & Feel:** Need accurate data, better insights
- **Hear:** Feedback from management
- **See:** Multiple data sources
- **Say & Do:** Collect statistics manually
- **Pain:** Data errors, delays
- **Gain:** Faster decisions, automation



2.3 Brainstorming

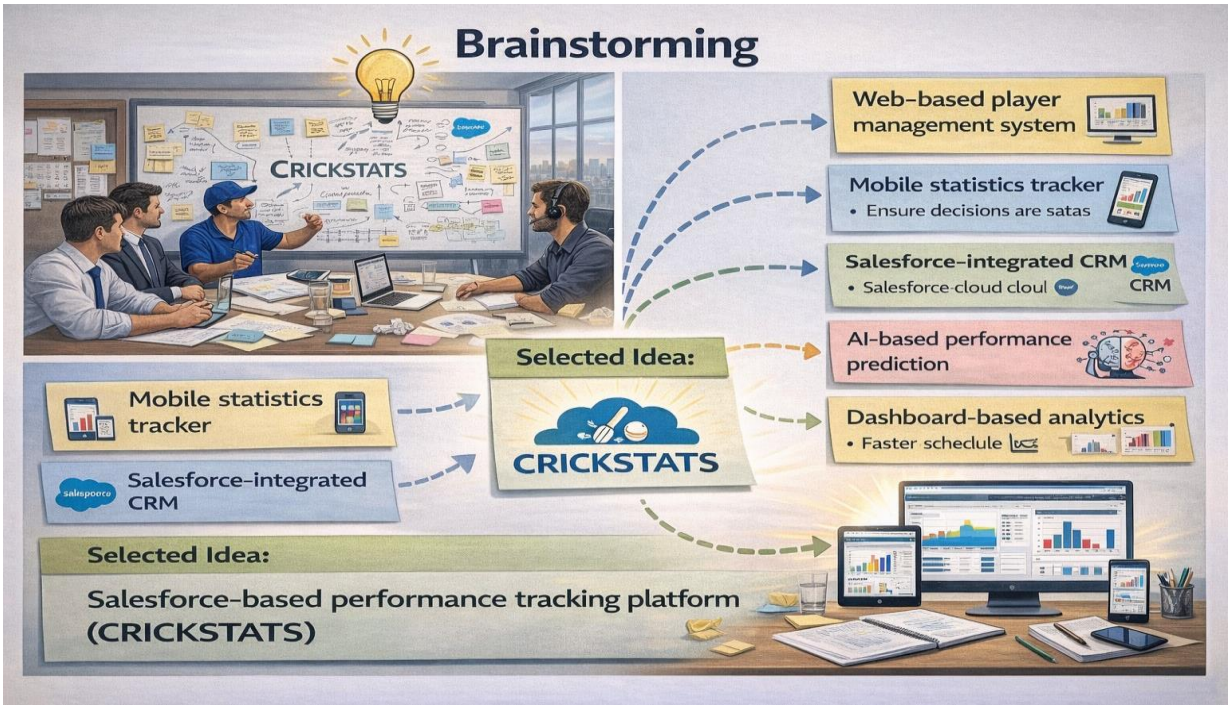
Brainstormed ideas

During the brainstorming phase, several ideas were considered for solving the problem of managing cricket performance data.

Some of the ideas included:

- Developing a web-based player management system
- Creating a mobile application for tracking cricket statistics
- Building a Salesforce-integrated CRM system
- Using artificial intelligence to predict player performance
- Designing dashboards for visual performance analysis

After evaluating all these ideas, the selected solution was to develop a Salesforce-based performance tracking platform called CRICKSTATS.This solution was chosen because Salesforce provides powerful tools for data management, automation, and analytics.



Brainstorming is an important step in the project development process where different ideas and solutions are discussed to solve a particular problem. In the CRICKSTATS project, brainstorming was conducted to identify possible ways to improve the management and tracking of cricket player performance data.During this phase, the team analyzed the problems faced in traditional cricket performance management,

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The customer journey describes the steps followed by users while interacting with the CRICKSTATS system.

The main stages include:

1. The user logs into the Salesforce system.
2. The user enters player details such as name, role, and team information.
3. The user records match performance statistics such as runs scored and wickets taken.
4. The system automatically updates player statistics using automation features.
5. Reports and dashboards are generated to visualize performance data.
6. Coaches and analysts review the reports and make decisions about player selection and team strategy.

This process ensures that all performance data is managed efficiently within the system.

3.2 Solution Requirements

Functional Requirements

Functional requirements describe the specific functions that the system must perform.

The CRICKSTATS system must provide the following functionalities:

- Player registration and profile management
- Match performance data entry
- Automatic calculation of player statistics
- Report generation for performance analysis
- Dashboard visualization for quick insights
- Automation of data updates using Salesforce Flows

These features allow users to manage cricket performance data effectively. Functional requirements describe the main functions and operations that the CRICKSTATS system must perform to manage cricket player performance effectively. The system should allow users to register and manage player details such as name, role, team, and other information. It should also enable users to enter match performance data including runs scored, wickets taken, balls faced, and overs bowled.

The CRICKSTATS application must automatically update player statistics using automation features such as Salesforce flows and roll-up summaries. The system should generate reports and dashboards to analyze player performance and provide visual insights. Additionally, the system should support easy navigation and data entry through the Salesforce Lightning interface so that coaches, managers, and analysts can efficiently track and manage cricket performance data.

The system should also provide the functionality to record match performance details. Users should be able to enter statistics such as runs scored, wickets taken, overs bowled, balls faced, and other match-related data. This information helps track the performance of players across different matches.

Another important requirement is the automatic calculation of player statistics. The CRICKSTATS system should use Salesforce features such as roll-up summary fields and automation flows to automatically calculate total runs, total wickets, and other performance metrics. This reduces manual calculations and ensures accuracy in performance analysis.

The system should also generate reports that allow users to analyze player performance over time. These reports help coaches and analysts compare player statistics and identify performance trends. In addition, dashboards should be available to visually represent player performance through charts and graphs, making it easier for users to understand the data.

The CRICKSTATS application should provide secure access to authorized users only. Role-based access control ensures that only permitted users can view or modify player data. This helps maintain data privacy and security within the system.

Finally, the system should support easy data entry and navigation through the Salesforce Lightning interface. A user-friendly interface allows coaches, managers, and analysts to efficiently manage player records and match performance data without requiring technical expertise.Overall, these functional requirements ensure that the CRICKSTATS system provides a reliable and efficient platform for tracking, managing, and analyzing cricket player performance.

The system should also include reporting and dashboard features. These features allow users to generate reports and view graphical representations of player statistics. Coaches and team managers can use these insights to evaluate player performance and make better decisions for team selection.

Security and accessibility are also key requirements. The application must ensure that only authorized users can access or modify important data. Role-based access control should be implemented to protect sensitive information.Overall, these solution requirements ensure that the CRICKSTATS system is efficient, reliable, and capable of managing cricket player performance data effectively.

4.2 Proposed Solution

Description

CIRCKSTATS provides:

- Custom objects for Player and Match Performance data management
- Centralized database to store all player statistics
- Master–detail relationships to connect player and match records
- Roll-up summaries to automatically calculate runs, wickets, and averages
- Automation using Salesforce Flows to update statistics automatically
- Reports and dashboards for performance analysis and visualization
- Secure access control for different users like coaches and managers
- Real-time tracking of player performance
- Easy data entry through Salesforce Lightning interface
- Scalable system that can support multiple teams and matches.

Key Features

- Centralized database for storing player and match performance data
- Real-time player statistics tracking
- Secure access to player information
- Automated calculation of player performance metrics
- Interactive dashboards and reports for analysis
- Easy data entry using Salesforce Lightning interface
- Scalable system that can support multiple teams and players
- Reduced manual work through automation.

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- Secure access control for different users like coaches and managers
- Real-time tracking of player performance
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- Scalable system that can support multiple teams and matches.

The proposed solution for this project is to develop a cloud-based cricket player performance tracking system called **CRICKSTATS** using the Salesforce platform. This system is designed to replace the traditional manual method of maintaining cricket statistics with an automated and centralized digital solution.In many cricket teams, player performance data such as runs scored, wickets taken, overs bowled, and match results are recorded manually in notebooks or spreadsheets. This approach often leads to errors, data loss, and difficulty in analyzing player performance. To overcome these issues, the CRICKSTATS application provides a structured platform where all player and match data can be stored and managed efficiently.

The system uses **Salesforce custom objects** such as **Player** and **Match Performance** to store player details and match statistics. These objects are connected using **master-detail relationships**, which allow the system to maintain proper data relationships and ensure data integrity. By using **roll-up summary fields**, the application automatically calculates player statistics such as total runs, total wickets, and average performance.

The proposed solution also includes **automation using Salesforce Flows**, which helps update player statistics automatically whenever new match data is entered into the system. This reduces manual work and ensures that the data remains accurate and up-to-date.In addition, the system provides **reports and dashboards** that allow coaches, team managers, and analysts to visualize player performance and identify trends. These analytical tools help users compare player statistics and make informed decisions regarding team selection and strategy.

Overall, the CRICKSTATS system provides a **secure, scalable, and efficient solution** for managing cricket player performance data. By using Salesforce technology, the application improves data accuracy, simplifies performance tracking, and supports better decision-making for cricket teams.

The CRICKSTATS application provides a comprehensive platform to track and manage cricket player performance efficiently. It enables users such as coaches, team managers, and analysts to store and analyze detailed match statistics in a centralized system.

The system provides custom objects like **Player** and **Match Performance** to record player information and

match statistics. It also includes **master-detail relationships** and **roll-up summaries** to automatically calculate player performance metrics such as total runs, wickets, and average scores.

CRICKSTATS Provides – Points

- Custom objects for **Player** and **Match Performance** data management
- **Centralized database** to store all player statistics
- **Master–detail relationships** to connect player and match records
- **Roll-up summaries** to automatically calculate runs, wickets, and averages
- **Automation using Salesforce Flows** to update statistics automatically
- **Reports and dashboards** for performance analysis and visualization
- **Secure access control** for different users like coaches and managers
- **Real-time tracking** of player performance
- **Easy data entry** through Salesforce Lightning interface
- **Scalable system** that can support multiple teams and matches

The CRICKSTATS system provides:

- Player Object
- Match Performance Object
- Master-Detail Relationship
- Roll-Up Summary Fields
- Automated Flows
- Interactive Dashboards

This system stores all cricket performance data in a centralized database and helps users analyze player performance easily.

CRICKSTATS also provides **automation using Salesforce flows**, which helps update player statistics automatically whenever new match data is entered. In addition, the system includes **interactive dashboards and reports** that allow users to visualize player performance and generate insights for team management.

Overall, CRICKSTATS provides a secure, scalable, and efficient solution for managing cricket player performance and improving data-driven decision making in cricket teams.

The application is developed using the Salesforce platform, which provides a secure and scalable environment for managing data. Custom objects such as **Player**, **Match**, and **Performance** are created to store important information about players and their match statistics. These objects are connected through relationships to ensure that the data is organized and properly linked. This structure helps users quickly access confusion

1.1 CRICKSTATS

provides:

- Custom objects (Player, Match Performance)
- Master-detail relationships
- Roll-up summaries
- Lightning App
- Automated flows
- Interactive dashboards

Key Features

- Centralized database
- Live statistics
- Secure accuracy
- Scalable design
- Easy reporting

The system is built using the Salesforce platform, which provides powerful tools for data management, automation, and reporting. Custom objects are created to store player details and match information, while relationships between these objects ensure that the data remains properly connected. This helps maintain an organized database and allows users to quickly retrieve information whenever required.

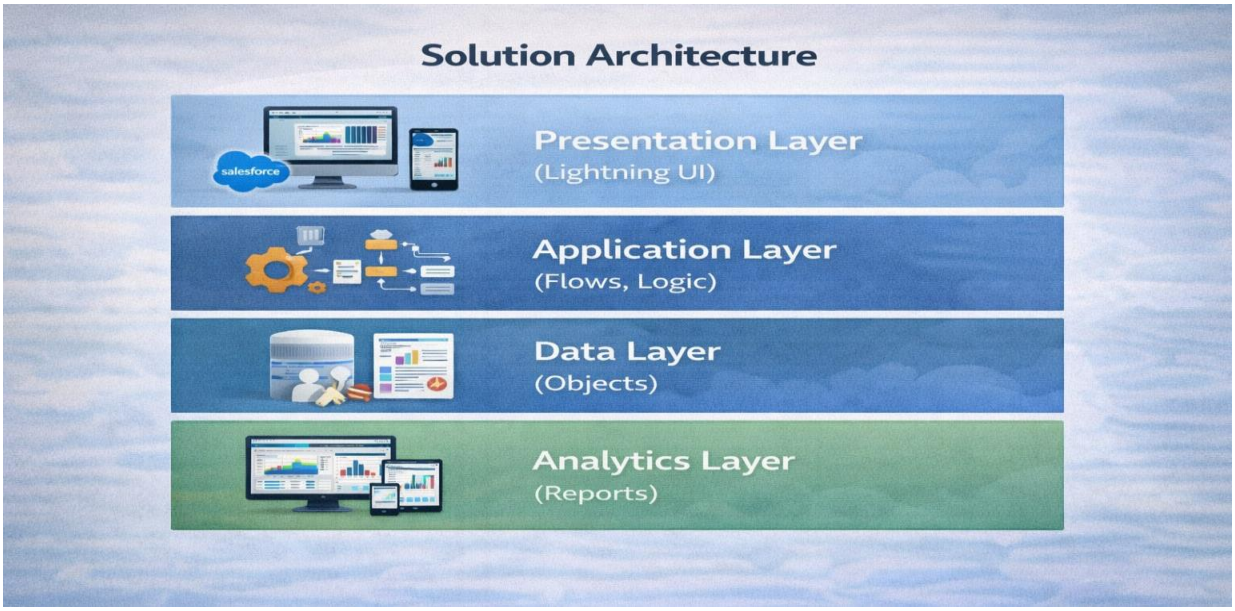
KEY FEATURES

- Real-time player statistics tracking
- Secure access to player information
- Automated calculation of player performance metrics
- Interactive dashboards and reports for analysis
- Easy data entry using Salesforce Lightning interface
- Scalable system that can support multiple teams and players
- Reduced manual work through automation.

4.3 Solution Architecture

Architecture Layers

1. **presentation Layer** – Salesforce Lightning UI for entering and viewing player data.
2. **Application Layer** – Salesforce Flows and logic for processing and automation.
3. **Data Layer** – Custom Objects (Player and Match Performance) to store data.
4. **Analytics Layer** – Reports and Dashboards for performance analysis.



5.PROJECT PLANNING & SCHEDULING

5.1Project Planning

- Agile-based execution
- Object creation → automation → reporting
- Sprint-wise development

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS system.	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Player and Match Performance objects so that player and match data can be stored systematically.	3	High	Member2
Sprint-2	Navigation	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage records within the application.	3	High	Member3
Sprint-2	UI Design	USN-4	As a coach, I want a dedicated Lightning App so that all performance tracking features are accessible from one dashboard.	3	High	Member4

Sprint-2	Data Structure	USN-5	As an admin, I want to create player and match fields so that complete performance details can be captured accurately.	5	Medium	Member1
Sprint-2	Data Structure	USN-6	As a user, I want customized page layouts so that only relevant fields are displayed during data entry.	5	High	Member2
Sprint-2	Automation	USN-7	As a system, I want to validate players and match inputs so that incorrect data entries are prevented.	3	High	Member4
Sprint-3	Automation	USN-8	As a system, I want automated flows to update player statistics so that performance data remains accurate.	5	High	Member1

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	**Total Story Points**	**Duration**	**Sprint Start Date**	**Sprint End Date (Planned)**	**Story Points Completed (as on Planned End Date)**
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	

2. Project Development Phase:

Project Flow

Milestone 1 : Creation of developer account

Milestone 2 :Object Creation

Milestone 3 :Tabs

Milestone 4 :The Lightning App

Milestone 5 :Fields & Relationships

Milestone 6 :Roll-up Summary

Implementation

<u>Milestone 1 - Salesforce developer account creation</u>

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

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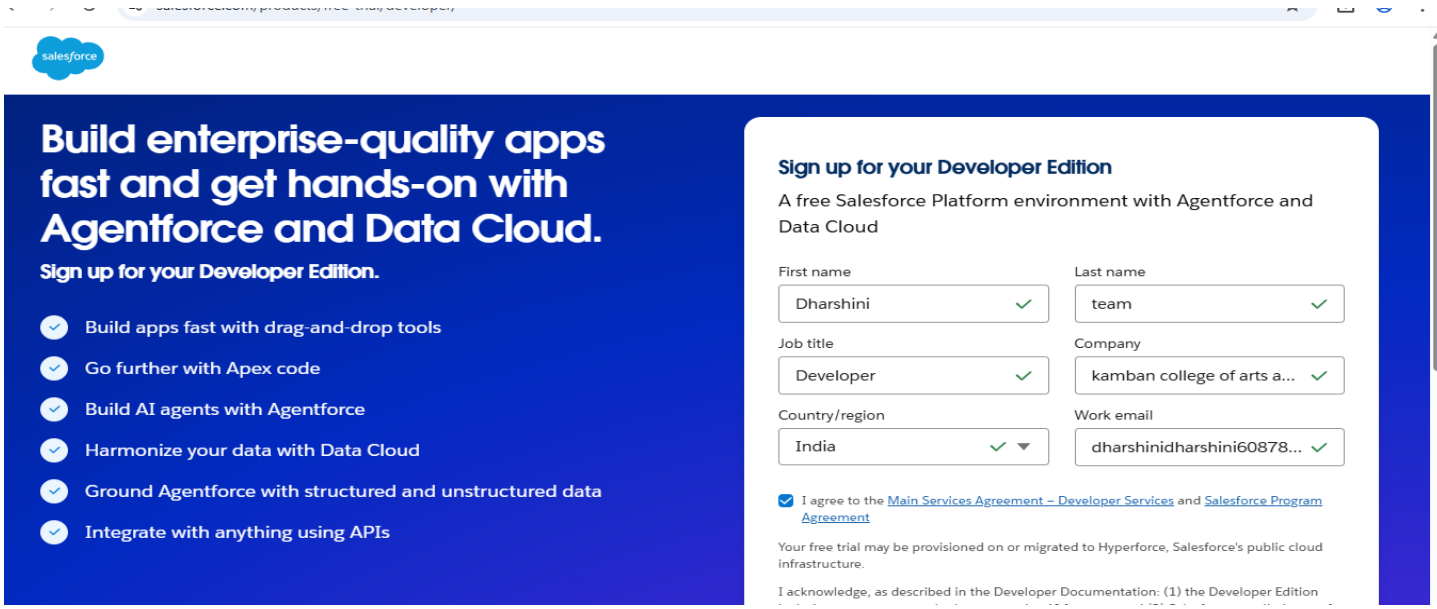
Go to Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :

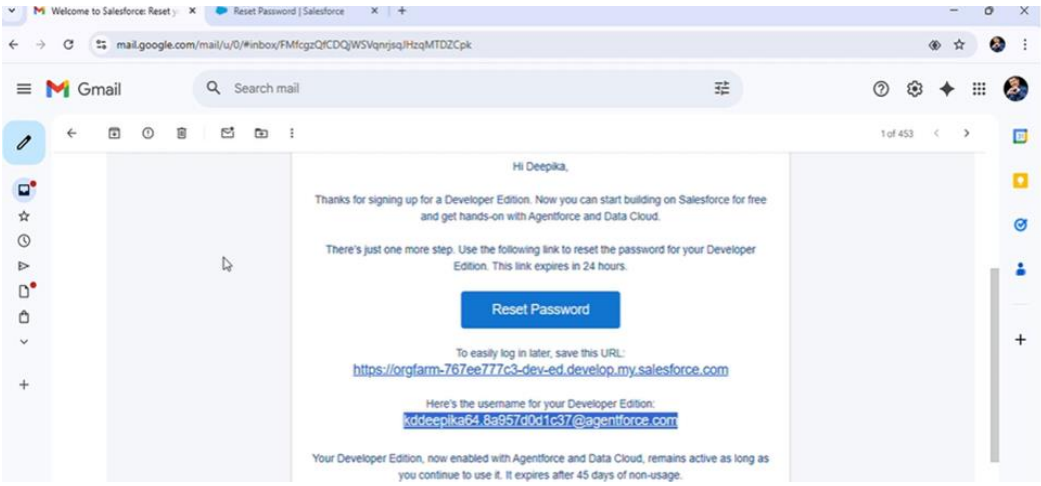


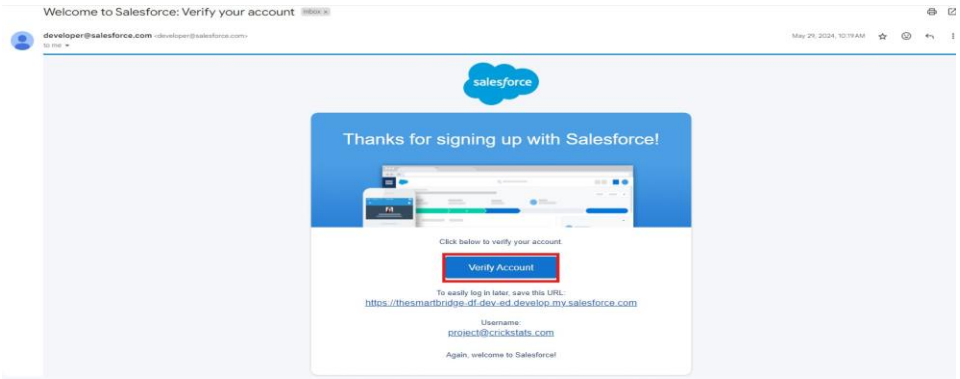
1) First name & Last name
2) Email
3) Role : Developer
4) Company : College or Company Name
5) County : India
6) Postal Code : pin code
7) Username : should be a combination of your name and company This need not be an actual email id, you can give anything in that format: username@organization.com

Click on sign me up after filling these.

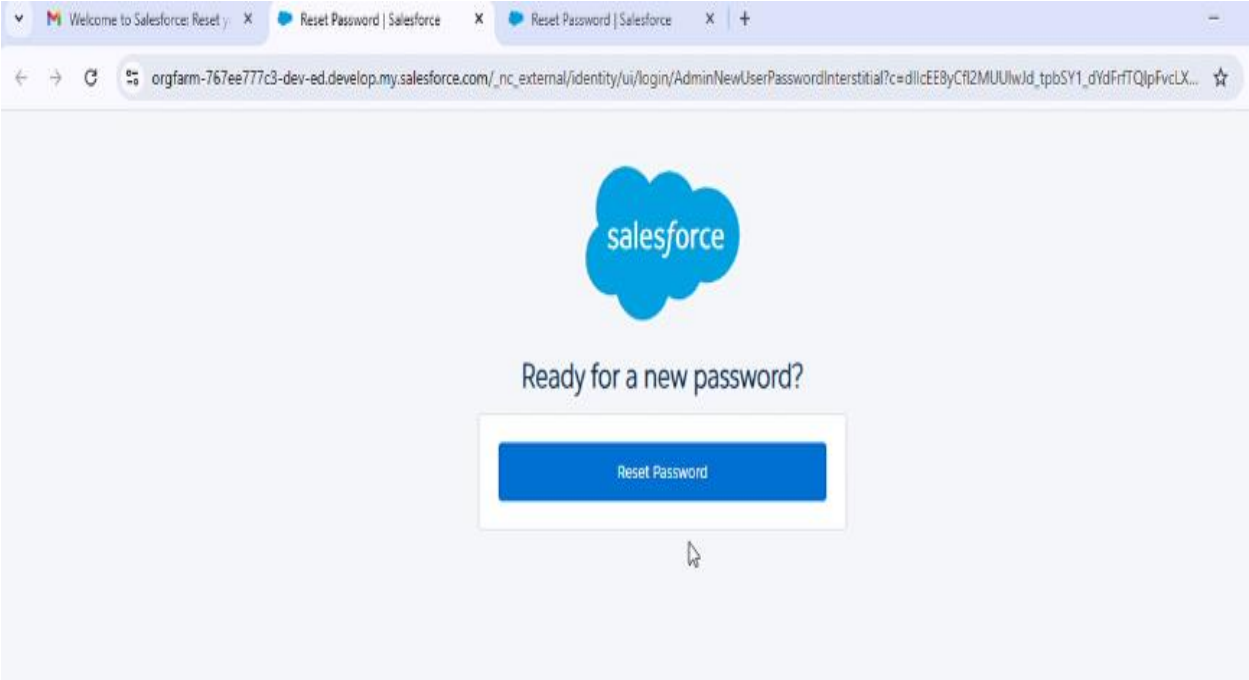
Activity 2: Account Activation :

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.
2. Click on Verify Account.

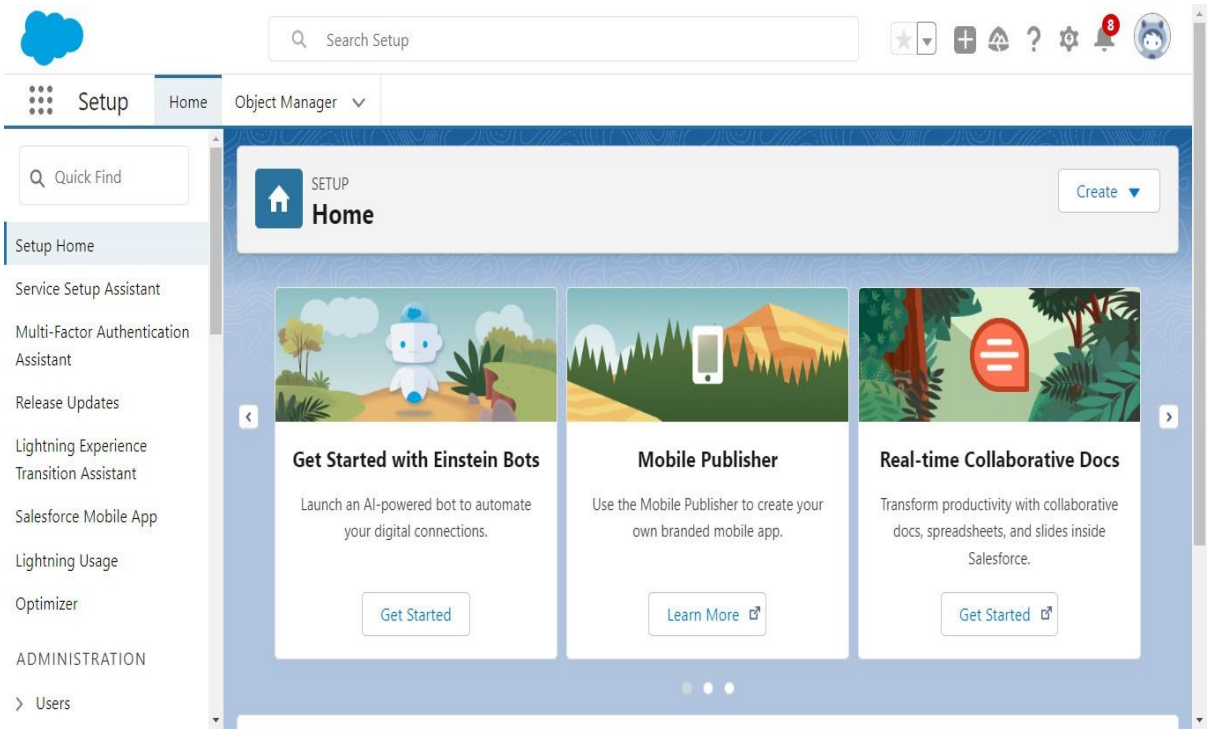




3.Give a password and answer a security question and click on change password.



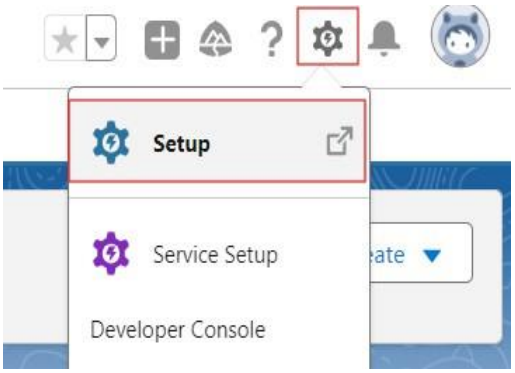
3. Then you will redirect to your salesforce setup page.



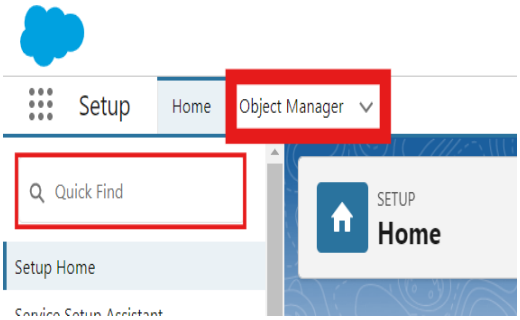
<u>Milestone 2 - Object Creation</u> Activity

1: Player Object

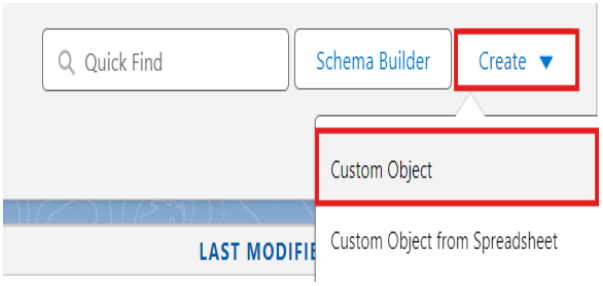
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.



2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.



3. On the Object Manager page, click Create | Custom Object.



1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
    - **Label:player**
    - **Plural labels:** Players

    Enter Record label and Format

    Record name: Play

    Player name Data Type**:** Text
    - **In Optional features :** check the boxes for Allow Reports allow low Activities | Track Field History.
    - **Search status:ch**eck the box for Allow Search.
    - Save

Activity 2: Match Performance Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
    - **Label**: Match Performance
    - **Plural Label**: Match Performances
    - Enter Record Name Label and Format

        Record Name: Customer Account Name

        Data Type**:** Text
    - **In Optional features:** check the boxes for Allow Reports | Allow Activities | Track Field History.
    - **Search Status:** check the box for Allow Search.
    - Save.

<u>Milestone 3 - Tabs</u>

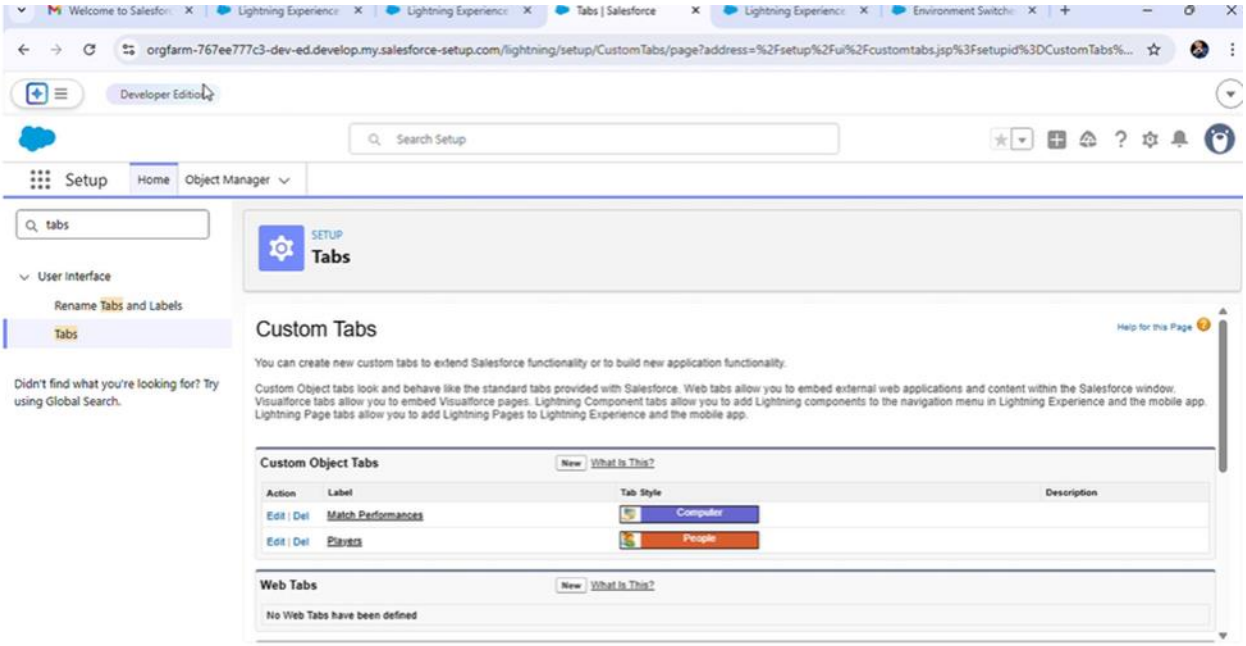
Activity 1: Custom tabs creation

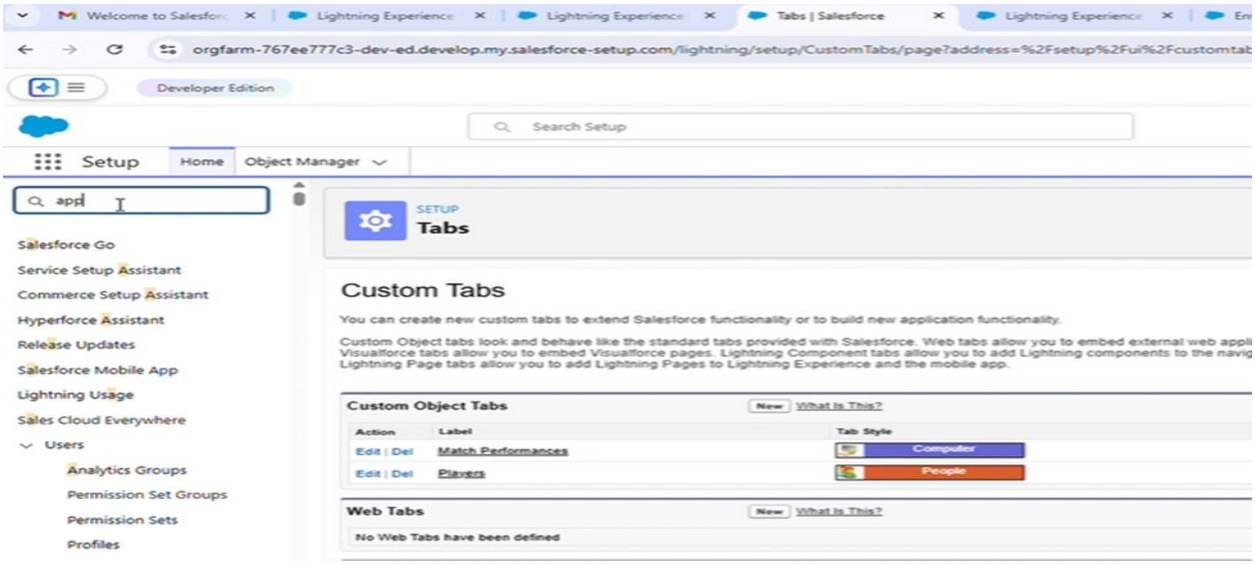
1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New

(under custom object tab)

2. Select Object (Player) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab.
3. Make sure that the Append tab to users' existing personal customizations is checked.
4. Click save.

Repeat the same steps for Match Performances





Milestone 3 focuses on the creation of **custom tabs** in the CRICKSTATS application. Tabs play an important role in improving the navigation and usability of the system. By creating tabs for each object, users can easily access and manage records related to players and match performances.

In this phase, custom tabs are created for the **Player** object and the **Match Performance** object in Salesforce. These tabs allow users such as coaches, managers, and analysts to quickly open the required sections of the application and perform operations like adding, editing, or viewing records.

The tab creation process begins in the Salesforce Setup page by navigating to the **Tabs section** and selecting the option to create a new custom object tab. After selecting the required object, a suitable tab style is chosen to represent the object visually in the navigation bar. The tab is then assigned to user profiles so that authorized users can access it within the CRICKSTATS application.

Once the tabs are created, they are added to the application navigation menu. This allows users to easily switch between different sections of the system, such as viewing player profiles or recording match performance statistics. Proper tab organization also improves the overall user experience by making the interface clean and structure.

Milestone 3 focuses on the system design and planning phase of the CRICKSTATS project. In this stage, the overall structure of the application is designed to ensure that the system functions efficiently and meets the project requirements. The main objective of this milestone is to create a clear blueprint for developing the system in the upcoming stages.

During this phase, different design components such as system architecture, data flow diagrams, and database structure are prepared. These designs help in understanding how data moves within the system and how different modules interact with each other. The project team identifies the main entities required for the system, such as Player, Match, and Performance, and defines the relationships between them.

The database structure is carefully designed to store all the necessary cricket statistics in an organized manner. Fields are identified for each object to capture important details like player name, team name, runs scored, wickets taken, and match details. This structured data model helps maintain accuracy and improves the efficiency of the system.

User interface planning is also carried out during this milestone. The team designs simple and user-friendly screens that allow users to easily enter player information and match performance data. Proper planning of page layouts and navigation ensures that the system will be easy to use for coaches, managers, and other users.

In addition, the technology stack and tools required for system development are finalized during this stage. Salesforce platform features such as custom objects, automation tools, and reporting features are selected to build the application.

Overall, Milestone 3 provides a detailed design and development plan for the CRICKSTATS system. This milestone ensures that the project has a clear structure and direction before moving into the actual development phase.

In Milestone 3, the CRICKSTATS project focuses on designing the data structure and system workflow. This stage is very important because it defines how the information will be stored, processed, and displayed in the application. A well-planned system design helps in developing an efficient and reliable application.

During this phase, the main entities required for the system are identified and structured properly. The primary objects created for the CRICKSTATS system include Player, Match, and Performance. Each object contains several fields that store important cricket data such as player name, age, team name, match date, runs scored, wickets taken, and other performance details. These objects are connected using relationships to ensure that the data is linked and managed effectively.

The system architecture is also planned in this stage to define how users interact with the application and how the data flows inside the system. Data Flow Diagrams (DFD) and process diagrams are designed to illustrate how information moves from the user to the database and how it is processed within the system.

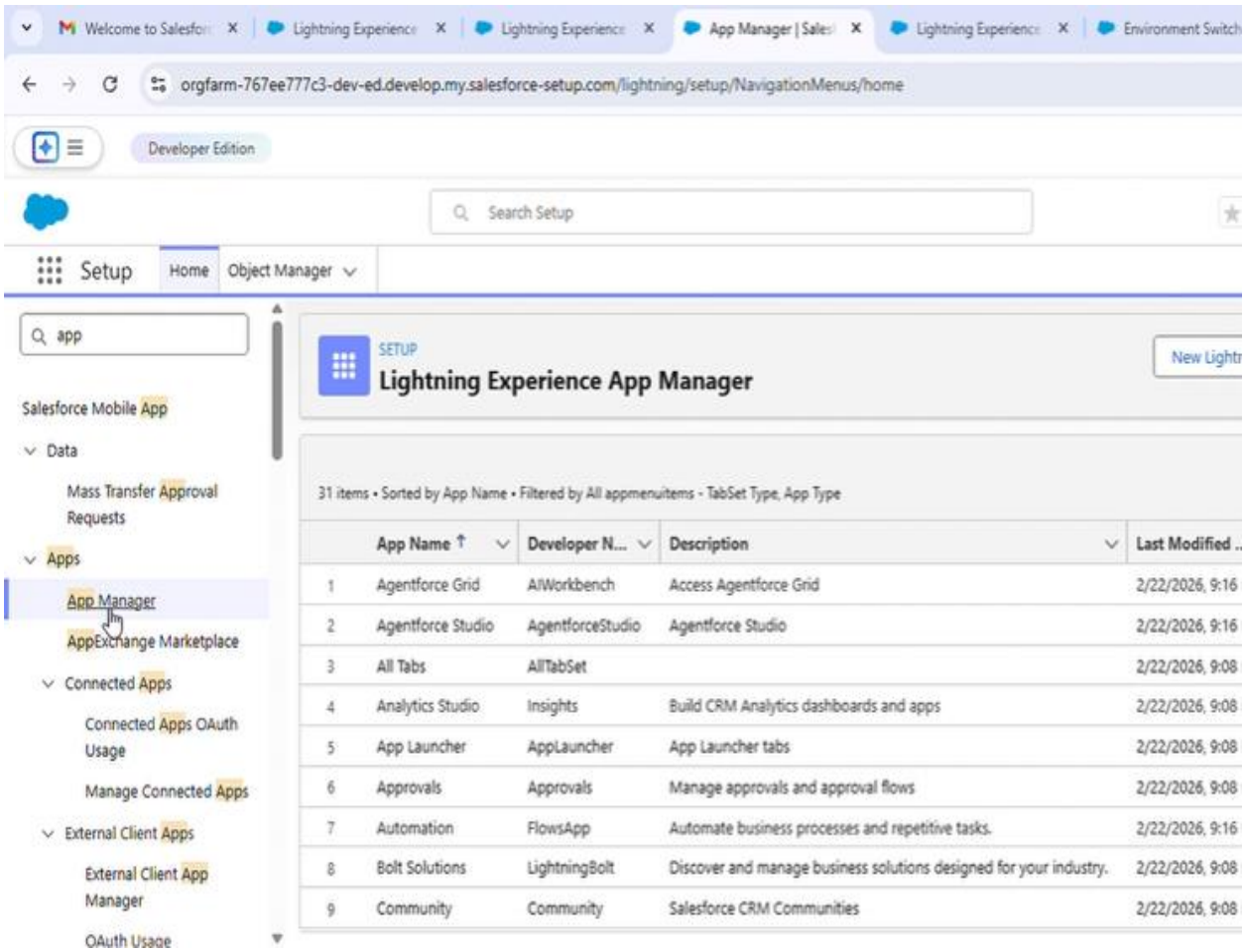
User interface design is another important activity in this milestone. Page layouts, record pages, and navigation structures are planned to make the application simple and user friendly. These designs help ensure that users can easily add, update, and view cricket statistics without confusion.

Security planning is also considered during this phase. Access permissions and user roles are defined so that only authorized users can modify important data. This helps maintain the integrity and security of the system.

Milestone 4 - The Lightning App

Activity 1: Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.

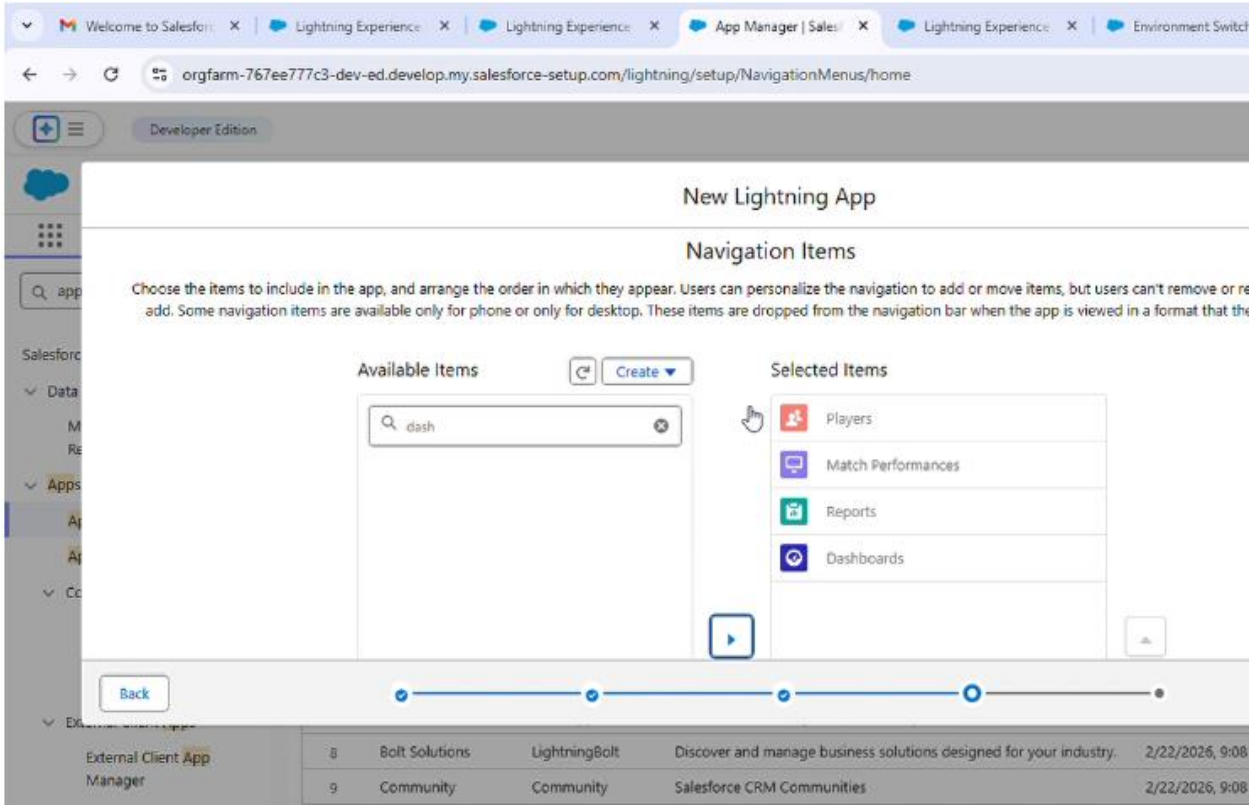


2. Fill the app name in app details and branding as follow App
   Name: CRICKSTATS
   Developer name: Auto Populates
   Add image optional (if you want to give any image you can otherwise not mandatory) -
   Add Primary color Hex or leave it to default.
   Then click Next.
3. In App options
   Navigation style: Standard navigation Setup
   experience: setup
   Supporters form factors : Desktop and phone Then
   click Next
4. In Utility items
   Utility Bar alignment : Default
   Then click Next.
5. Navigation items
   Select the created Custom Objects and required standard objects

   - Players
     Match Performances
     Reports
     Dashboards

Milestone 4 focuses on the development and integration of the CRICKSTATS system. In this stage, the system design created in the previous milestones is converted into a working application using the Salesforce platform. The main objective of this milestone is to build the core components of the system and ensure that all modules work together effectively.

During this phase, custom objects such as **Player**, **Match**, and **Performance** are created to store and manage cricket-related data. These objects are designed to maintain important information including player details, match records, runs scored, wickets taken, and other performance statistics. Proper relationships are also established between these objects to ensure accurate data linking and organization.

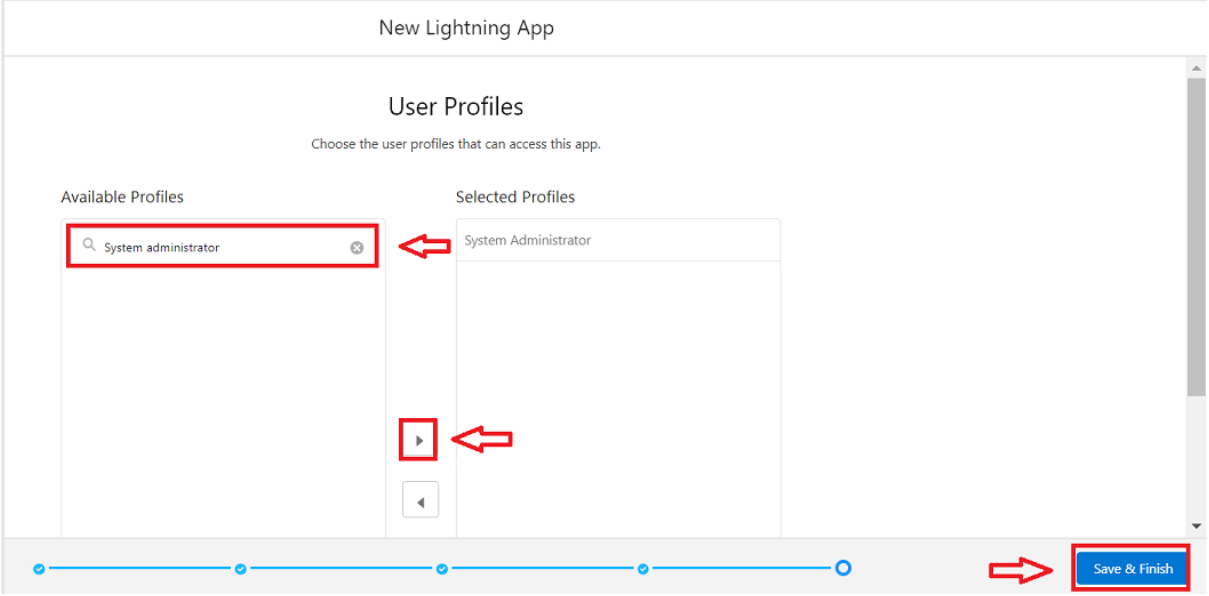


Milestone 4 focuses on creating **custom fields and establishing relationships between objects** in the CRICKSTATS project. This stage is important because it helps store detailed information about players and their match performances in a structured way.In this phase, different types of fields are created in the **Player** and **Match Performance** objects. These fields are used to capture important cricket statistics such as player name, team name, role, runs scored, wickets taken, match date, strike rate, and other performance details. Creating these fields allows the system to store accurate and organized data related to cricket matches.

After creating the required fields, relationships are established between objects. A relationship is used to connect one object with another so that related data can be easily accessed. In the CRICKSTATS project, a **relationship is created between the Player object and the Match Performance object**. This allows each match performance record to be linked to a specific player.

The relationship helps users track the performance of each player across different matches. Coaches, analysts, and managers can view the complete performance history of a player in a single place. This makes the system more efficient for analyzing cricket statistics.By completing Milestone 4, the CRICKSTATS application gains a well-structured data model where all player information and match performance details are properly connected. This improves data management, reporting, and analysis within the system.

6.To Add User Profiles: System Administrator

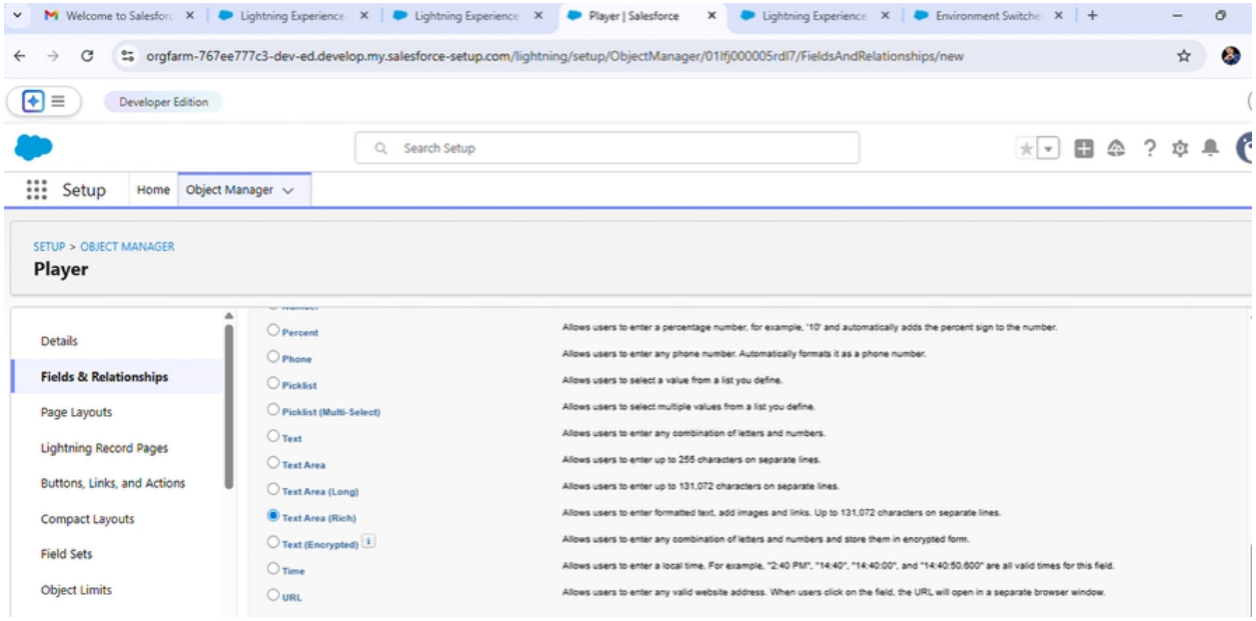


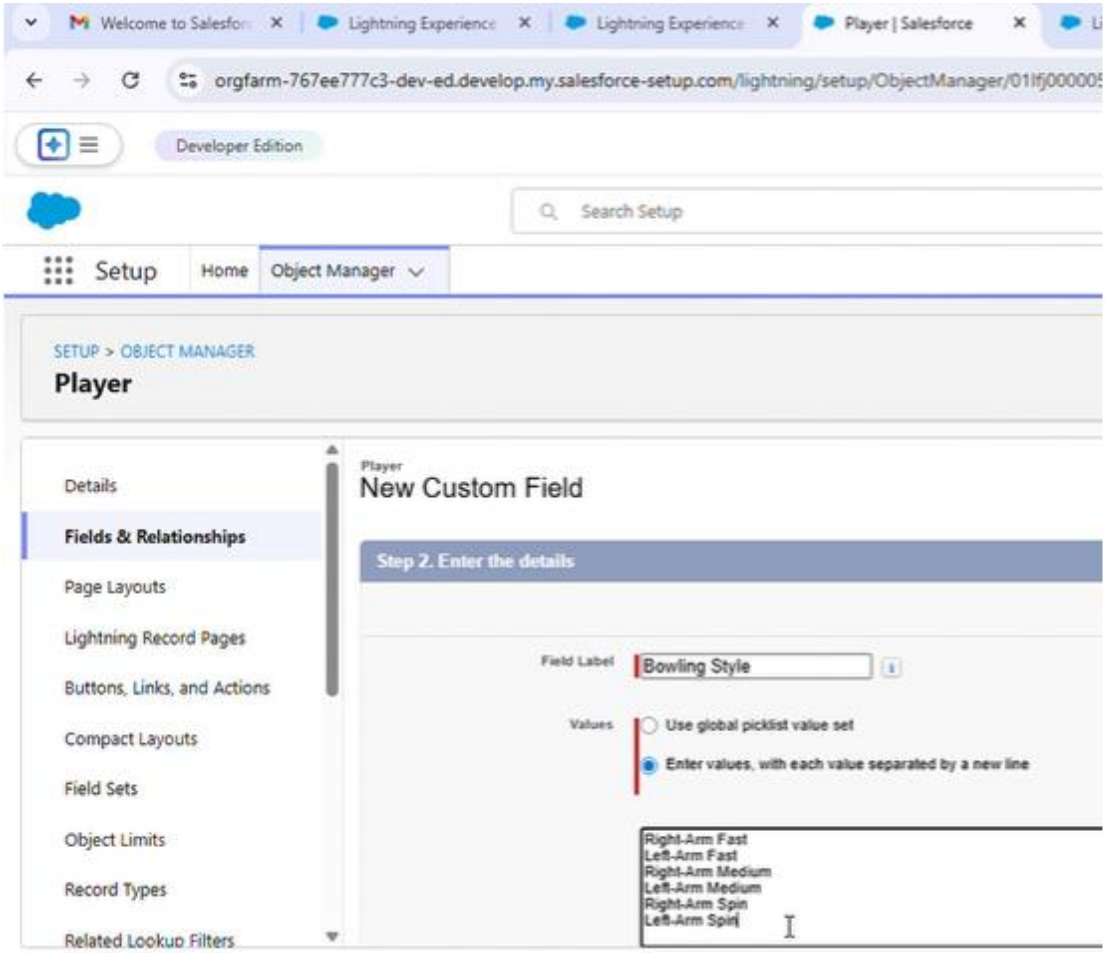
6. Click Save & Finish.

<u>Milestone 5 :Fields & Relationships</u>

Activity 1: Create Custom field for Player object

a. In your salesforce org, click gear icon on the top left and select Setup to open Setup.
b. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
c. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**
d. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
e. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist.
f. Next, choose a data type as Text. Choosing a data type helps you format the field input.


Welcome to Salesforc
Lightning Experience
Lightning Experience
Player | Salesforce
Lightning Experience
Environment Switche
orgfarm-767ee777c3-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01Ifj000005rdl7/FieldsAndRelationships/new
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Player
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Percent
Allows users to enter a percentage number, for example, '10' and automatically adds the percent sign to the number.
Phone
Allows users to enter any phone number. Automatically formats it as a phone number.
Picklist
Allows users to select a value from a list you define.
Picklist (Multi-Select)
Allows users to select multiple values from a list you define.
Text
Allows users to enter any combination of letters and numbers.
Text Area
Allows users to enter up to 255 characters on separate lines.
Text Area (Long)
Allows users to enter up to 131,072 characters on separate lines.
Text Area (Rich)
Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.
Text (Encrypted)
Allows users to enter any combination of letters and numbers and store them in encrypted form.
Time
Allows users to enter a local time. For example, "2:40 PM", "14:40", "14:40:00", and "14:40:50.600" are all valid times for this field.
URL
Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.


Welcome to Salesforc
Lightning Experience
Lightning Experience
Player | Salesforce
orgfarm-767ee777c3-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01Ifj000005
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Player
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Player
New Custom Field
Step 2. Enter the details
Field Label
Bowling Style
Values
Use global picklist value set
Enter values, with each value separated by a new line
Right-Arm Fast
Left-Arm Fast
Right-Arm Medium
Left-Arm Medium
Right-Arm Spin
Left-Arm Spin

1. Click on next.

2. Data Type: **Text**

   Field Label: Player Name Field

   Name : Auto Populates

   Length :50

   Click on Next → Next → Save and new.

Repeat the Same steps for remaining fields

- Data Type: **Text**

  Field Label: Player Number

  Field Name : Auto Populates

  Length : 3

  Click on Next → Next → Save and new.

- Data Type : **Rich Text Area**

  Field Label : Profile Picture

  Field Name : Auto Populates

  Click on Next → Next → Save and new.

- Data Type : **Date**

  Field Label : Date of Birth

  Field Name : Auto Populates

  Click on Next → Next → Save and new.

- Data Type : **Picklist**

  Field Label : Role

  Field name: Auto Populates

  Value: Enter value, with each value separated by a new line.

  Batsman

  Bowler

  Batsman & Wicket Keeper All

Rounder

- Data type:**Picklist**

  Field Label : Batting Style

  Field Name : Auto Populates

  Value : Enter value, with each value separated by a new line.

  Right- Handed

  Left - Handed

  Click on Next → Next → Save and new.

- Data Type : **Picklist**

  Field Label : Bowling Style

  Field Name : Auto Populates

  Value : Enter value, with each value separated by a new line.

  Right-Arm

  Fast Left-Arm

  Fast

Functional Requirements (Milestone 5)

- The system should allow users to create and manage player records.
- The system should allow users to enter match performance details such as runs, wickets, and overs.
- The system should store player and match data securely in the Salesforce database.
- The system should automatically calculate player statistics using roll-up summary fields.
- The system should generate reports to analyze player performance.
- The system should display dashboards for visual representation of player statistics.
- The system should automate data updates using Salesforce Flows.
- The system should provide secure access for authorized users like coaches and managers.

Milestone 5 represents the final stage of the CRICKSTATS project where the system is fully implemented, tested, and prepared for deployment. In this phase, all the components of the application are integrated and verified to ensure that the system functions correctly according to the project requirements.

During this milestone, the CRICKSTATS application is implemented on the Salesforce platform using custom objects such as Player and Match Performance. These objects are used to store detailed information about cricket players and their match statistics. The system allows users to enter and manage player details, record match performance data, and maintain a structured database of all player records.

Automation features such as Salesforce Flows and roll-up summary fields are implemented to automatically update player statistics. Whenever match data is entered into the system, the application automatically calculates important performance metrics such as total runs, wickets taken, and match statistics. This automation reduces manual work and improves the accuracy of performance tracking.

Reports and dashboards are also created in this milestone to analyze player performance. These analytical tools help coaches, team managers, and analysts visualize player statistics and identify performance trends. Dashboards provide graphical representations such as charts and tables that make it easier to understand player performance data.

Testing is another important part of Milestone 5. The system is tested to ensure that all functionalities work correctly, including data entry, automation processes, report generation, and dashboard visualization. Any errors identified during testing are corrected to ensure smooth system operation.

Finally, the system is prepared for deployment so that users can access the application and start managing cricket player performance data effectively. With the successful completion of Milestone 5, the CRICKSTATS application becomes a fully functional platform that supports efficient data management, performance analysis, and better decision-making for cricket teams.

Click on Next → Next → Save and new.

- Data type:**Picklist**

  Field label:Nationality

  Field Name:auto papulates

  Value: Enter value, with each value separated by a new line.

  Australia
  Banglad
  esh
  England
  India
  New
  Zealand
  Pakistan
  South
  Africa
  Sri Lanka
  West
  Indies
  Afghanist
  an
  Zimbabw
  e Ireland
  Scotland
  Netherlan
  ds
  United Arab
  Emirates Nepal
  United States of America

  Click on Next → Next → Save and new.

- Data Type : **Picklist**

  Field Label : Team

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line

Chennai Super Kings (CSK)
Delhi Capitals (DC)
Kolkata Knight Riders (KKR) Mumbai Indians (MI)
Punjab Kings (PBKS) Rajasthan Royals (RR)
Royal Challengers Bangalore (RCB) Sunrisers Hyderabad (SRH)
Gujarat Titans (GT)
Lucknow Supe Giants (LSG)

Milestone 5 focuses on creating **reports and dashboards** in the CRICKSTATS project to analyze player performance and match statistics effectively. Reports help in organizing and presenting the stored data in a meaningful format, while dashboards provide a visual representation of the information for quick understang.

In this phase, reports are created based on the data stored in the **Player** and **Match Performance** objects. These reports display important details such as total runs scored by players, number of wickets taken, match performance statistics, and overall player records. Reports allow coaches, analysts, and team managers to easily track and evaluate player performance across different matches.

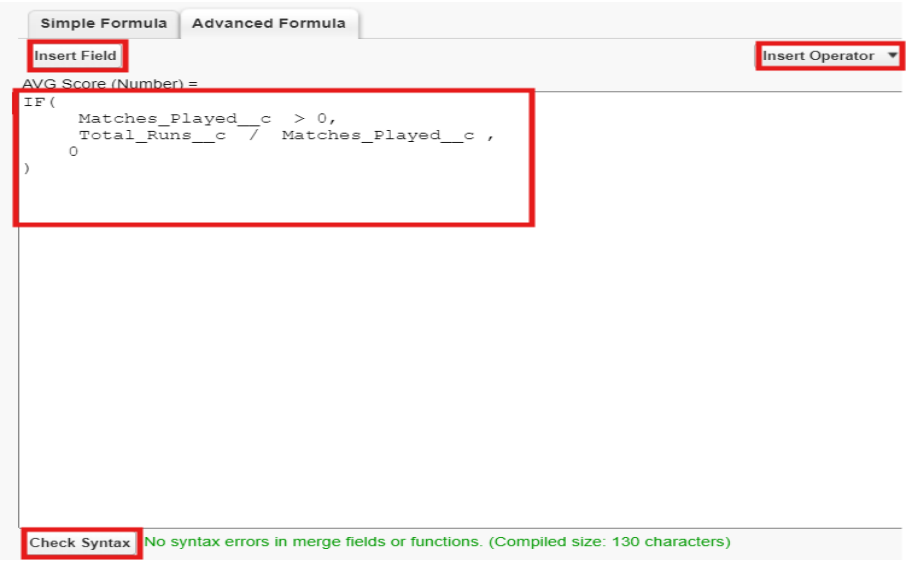
After creating reports, dashboards are designed to visually present the report data using charts, graphs, and summary components. Dashboards help users quickly understand performance trends, compare players, and identify key statistics without going through large amounts of raw data.The dashboard may include components such as **top performing players, total runs scored, wicket statistics, and match performance summaries**. These visual elements make the system more interactive and easier to interpret.

With the completion of Milestone 5, the CRICKSTATS system provides powerful reporting and visualization features. This enables users to make better decisions based on player statistics and match performance data, improving the overall analysis capability of the project.

Data Type : **Formula**

Field Label : AVG Score
Field Name : Auto
Populates Formula Return :
Number

Number:if(
Matches_Played_c > 0,
Total_Runs_c / Matches_Played_c ,
0
)



Click on Next → Next → Save & New.

- Data Type: Lookup Relationship
  Related To: Match Performance
  Field Label: Match Performance
  Field Name: Auto Populates

Click on Next → Next → Save.

Milestone 5 – Final Implementation and System Testing

Milestone 5 is the final phase of the CRICKSTATS project where the developed system is implemented, tested, and prepared for final submission. In this stage, all the modules and functionalities designed in the previous milestones are integrated to create a complete and working application. The goal of this milestone is to ensure that the system works efficiently and satisfies all the functional requirements of the project.During this phase, the CRICKSTATS application is fully developed using the Salesforce platform. Custom objects such as Player, Match, and Performance are created to store and manage the cricket statistics. These objects help organize important information such as player details, match records, runs scored, wickets taken, and other performance metrics. Relationships between these objects are also established to maintain proper data structure and improve data management within the system.

User-friendly interfaces are designed to allow users to easily enter and update information in the system. Forms and record pages are created so that team managers or coaches can add player details, record match performances, and view player statistics without any difficulty. Automation tools such as Salesforce flows are also implemented to automatically update player statistics when new match data is entered into the system. This helps reduce manual effort and improves data accuracy.

Testing is an important part of this milestone. Different types of testing such as functional testing and system testing are performed to ensure that the application works correctly. Each feature of the system is checked carefully to confirm that it performs according to the project requirements. Any issues or errors found during testing are fixed to ensure smooth system operation.

Reports and dashboards are also created in this stage to provide visual representation of player performance data. These tools allow users to analyze player statistics through charts and graphs, making it easier to understand performance trends and compare players.

Finally, the CRICKSTATS system is successfully implemented and ready for real-time usage. This milestone confirms that the application is fully functional, reliable, and capable of efficiently managing cricket player performance data. The successful completion of Milestone 5 marks the final step in the development of the project.

Field Name : Auto Populates
Click on Next → Next → Save and new.

1. Data Type : **Picklist**
   Field Label: Match Result Field
   Name : Auto Populates
   Value : Enter value, with each value separated by a new line
   WON
   LOST
   DRAW
   Click Next → Next → Save and new.

2. Data Type : **Number**
   Field Label: Runs Scored Field
   Name : Auto Populates
   Click on Next → Next → Save and new.

3. Data Type : **Number**
   Field Label: Balls Faced Field
   Name : Auto Populates
   Click on Next → Next → Save and new.

4. Data Type : **Number**
   Field Label: Fours
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

5. Data Type : **Number**
   Field Label: Sixes
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

6. Data Type : **Number**
   Field Label: Overs Bowled Field
   Name : Auto Populates
   Click on Next → Next → Save and new.

7. Data Type : **Number**
   Field Label: Wickets Taken
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

8. Data Type : **Number**
   Field Label: Runs Conceded
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

9. Data Type : **Number**
   Field Label: Maiden Overs
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

10. Data Type : **Number**
    Field Label: Stumpings
    Field Name : Auto Populates
    Click on Next → Next → Save and new

11. Data Type : **Number**
    Field Label: Catches
    Field Name : Auto Populates
    Click on Next → Next → Save and new.

12. Data Type : **Master - Detail Relationship**
    Related To : Player
    Field Label : Player
    Field Name : Auto Populates
    Click on Next → Next → Save.

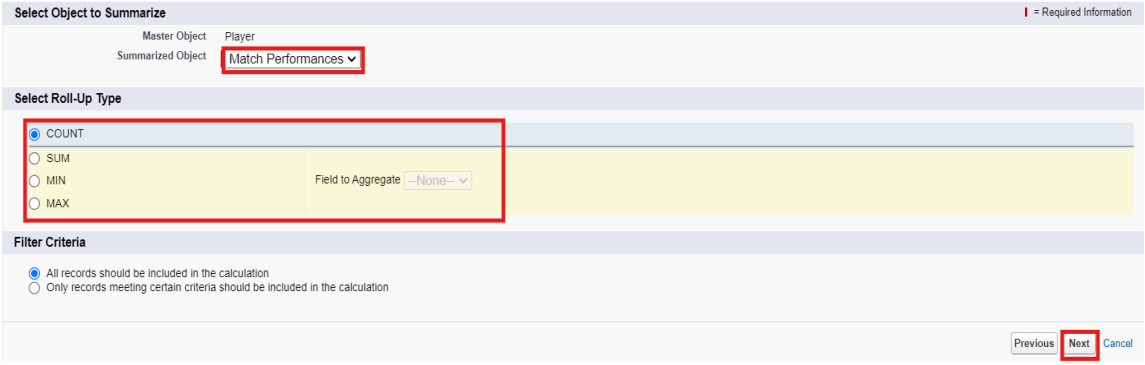
<u>Milestone 6 : Roll-up Summary</u>

Activity 1 : Create Roll-up Summary for Player object.

Note:- Roll-up Summary can only be created on parent object in Master- Detail Relationship.

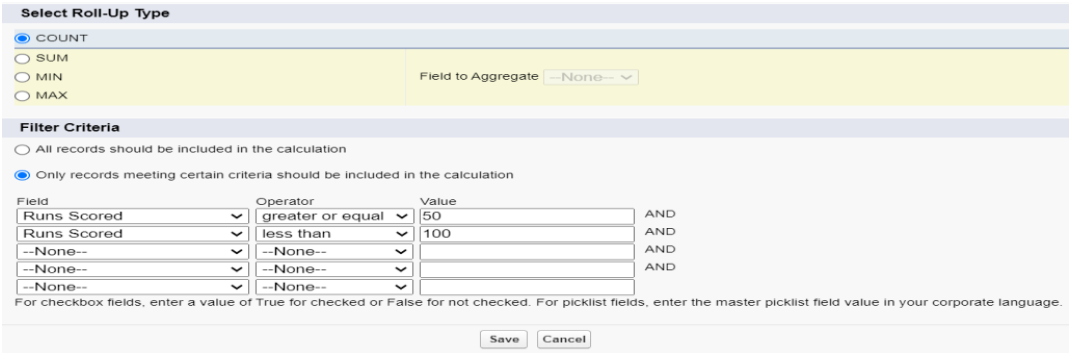
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created:**Player**

4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist. Click Next,
6. Choose a data type as Roll-up Summary. Click Next. Field
   Label: Matches Played
   Field Name : Auto Populates
   Summarized Object : Match Performances Select
   Roll-up Type : COUNT

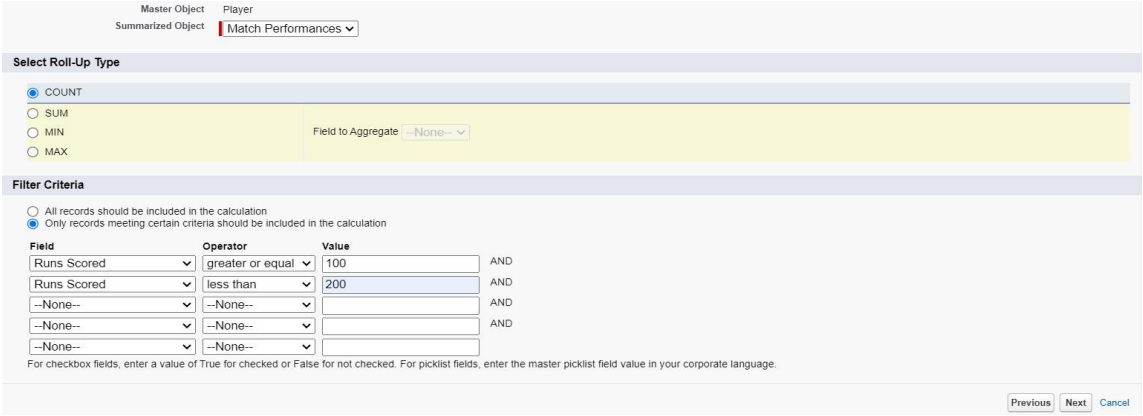


Filter criteria: All records should be included in the calculation.
Click on Next → Next → Save and New.

- Data Type : Roll-up Summary
  Field Label: Number of 50's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type: COUNT
  Filter Criteria : Only records meeting certain criteria should be included in the calculation.
  Click on Next → Next → Save and new.



- Data Type : Roll-up Summary
  Field Label: Number of 100's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : COUNT
  Filter Criteria : Only records meeting certain criteria should be included in the calculation.
  Click on Next → Next → Save and new.



- Data Type : Roll-up Summary
  Field Label: Total 4's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Fours
  Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total 6's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Sixes
  Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Runs
  Field Name : Auto Populates

Summarized Object : Match Performances
Select Roll-up Type : SUM
Field To Aggregate : Runs Scored
Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Overs
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Overs Bowled
  Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

- Data type: Roll-up Summary
  Field Label: Total Wickets
- Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Wickets Taken
  Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

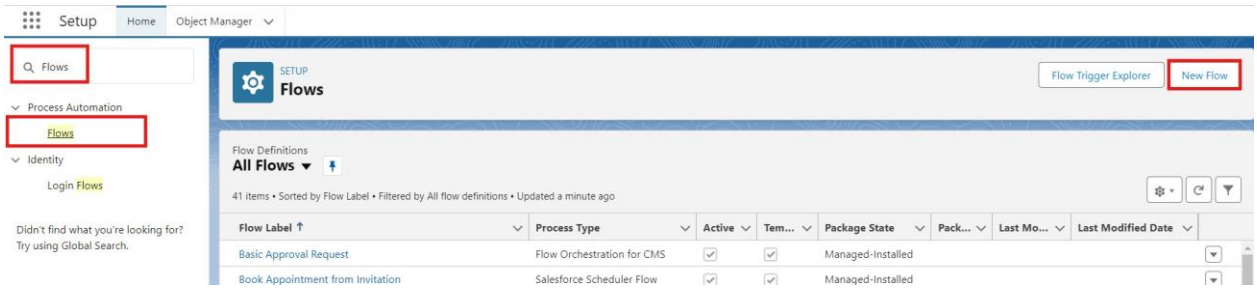
- Data Type : Roll-up Summary
  Field Label: Total Stumpings
  Field Name : Auto Populates
  Summarized object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Stumpings
  Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
  Field Label: Total Catches Field
  Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Catches
  Filter Criteria : All records should be included in the calculation. Click on Next → Next → Save and new.

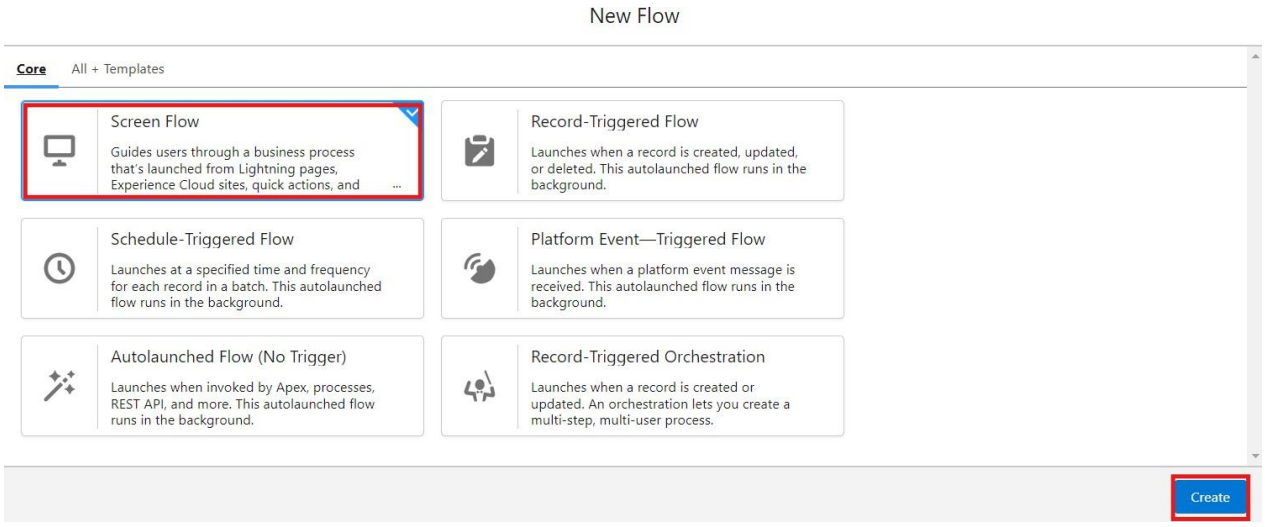
Milestone 7 : Flows Activity

Screen Flow

1. Go to setup, Type Flow in quick find box → Click on the Flow and Select the New Flow.

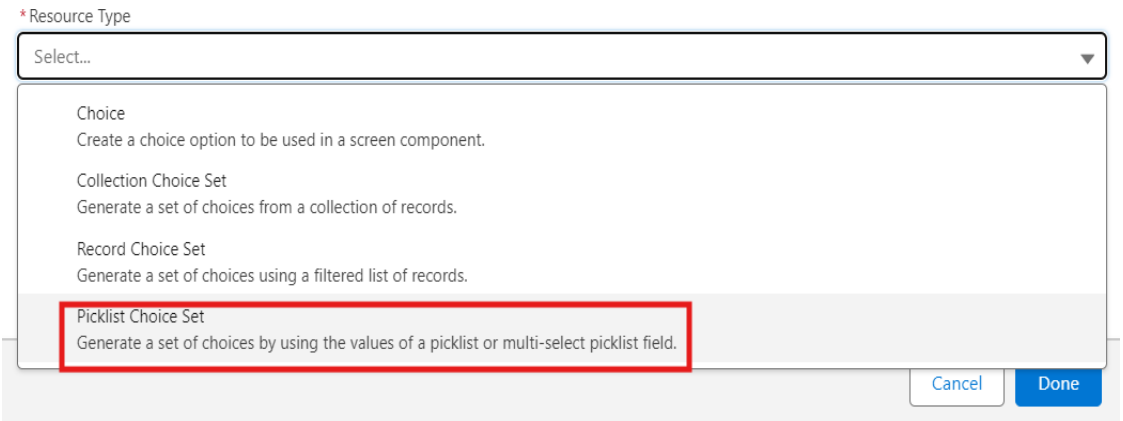


2. Select Screen Flow and click Create.

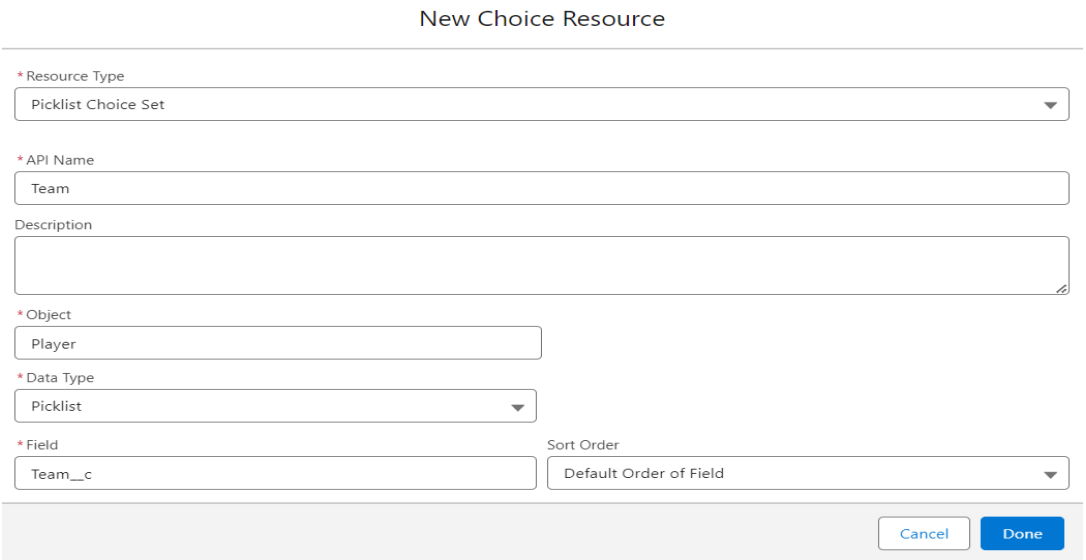


3. Click (+) Icon on the flow builder and Select **Screen Component.**
   Label : Players screen
   API : Players_screen
   Select the **Picklist** component and place it on the screen.
   Label : Team
   API : Team
   From Component Pane on the left side select **Component Type** : Picklist
   Data Type : Text
   Choice : Click New Choice Resources

Resource Type : Picklist Choice Set

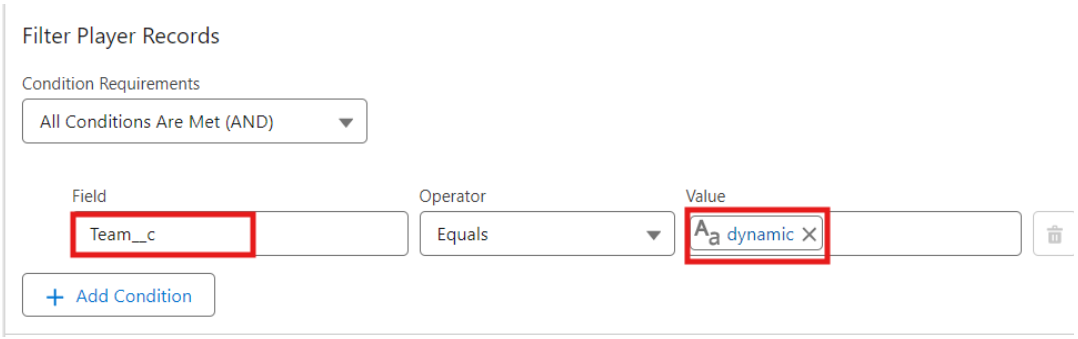


API Name: Team
Object : Player
Type : Pick list
Field : Team
Click Done.

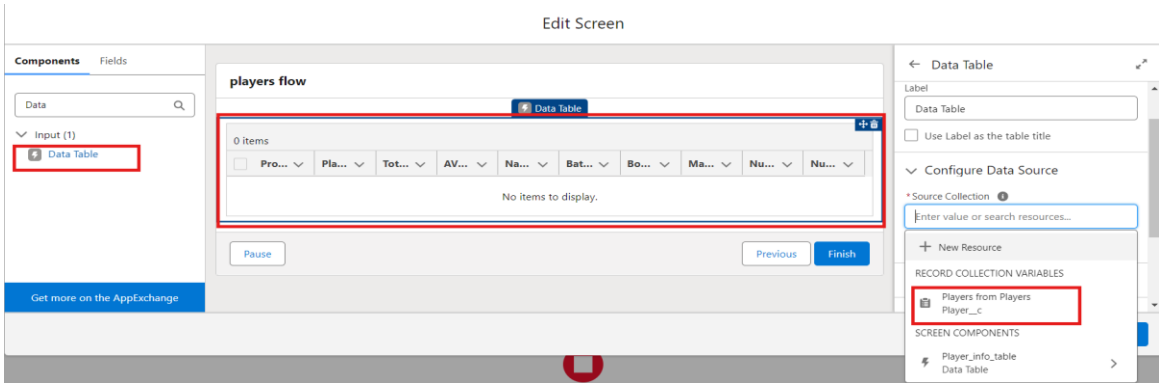


4. Click (+) Icon on the flow builder and Select **Get Records Component.**
5. **Label :** Players
   API Name : Players
   Object : Player
   Filter player records
   Field : Team
   Operator : Equals

Value : Click New Resource
Resource Type : Variable **API**
Name : Dynamic
Data Type : Text
Default Value : Screen component : Team (Picklist)



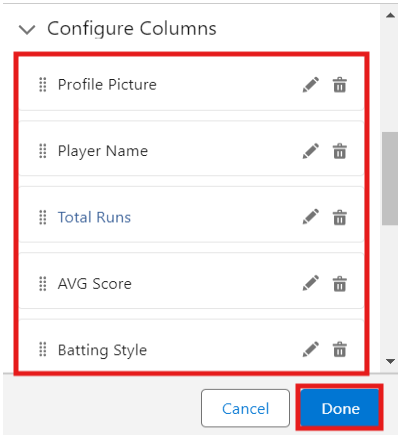
6. Click (+) Icon on the flow builder and Select **Screen Component.**
Label : Player_info
API Name : Player_info
From Component Pane on the left side select **Component Type :**Data table
API Name :Player_info_table



In Configure Columns : Add all the required Columns

Profile Picture | Player Name | Total Runs | AVG Score | Batting Style | Bowling Style | Matches Played | Number of 50's | Number of 100's

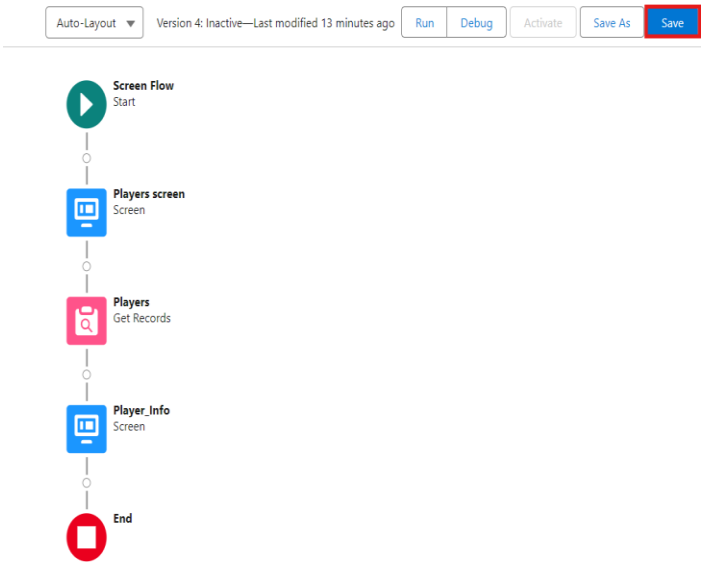
Click Done.

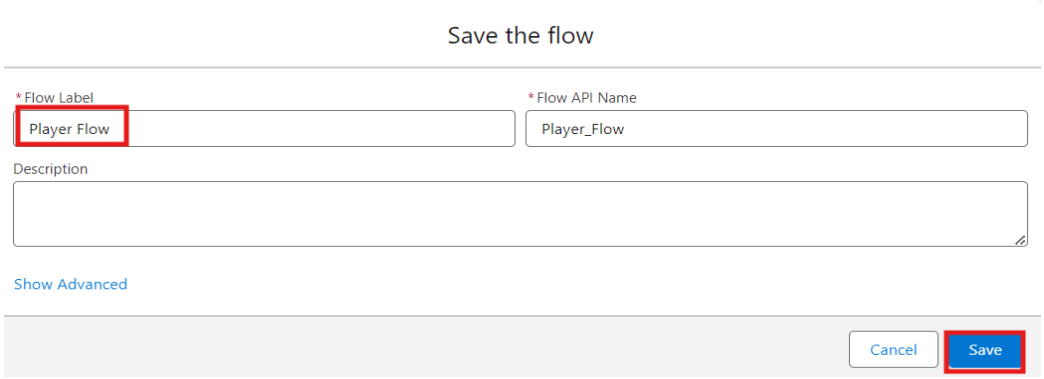


Click Save.

Flow Label : Player Flow

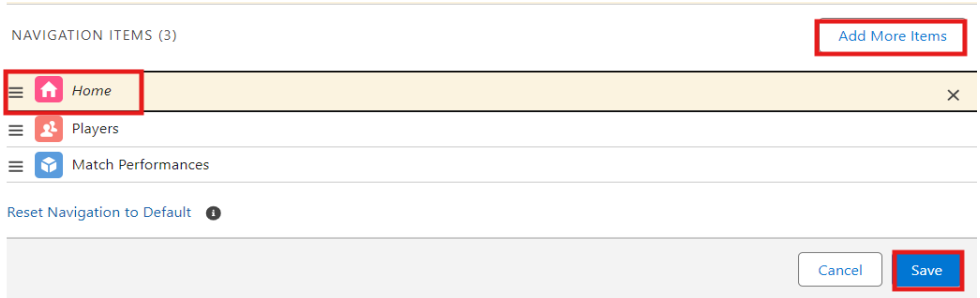
Flow API Name : Player_Flow



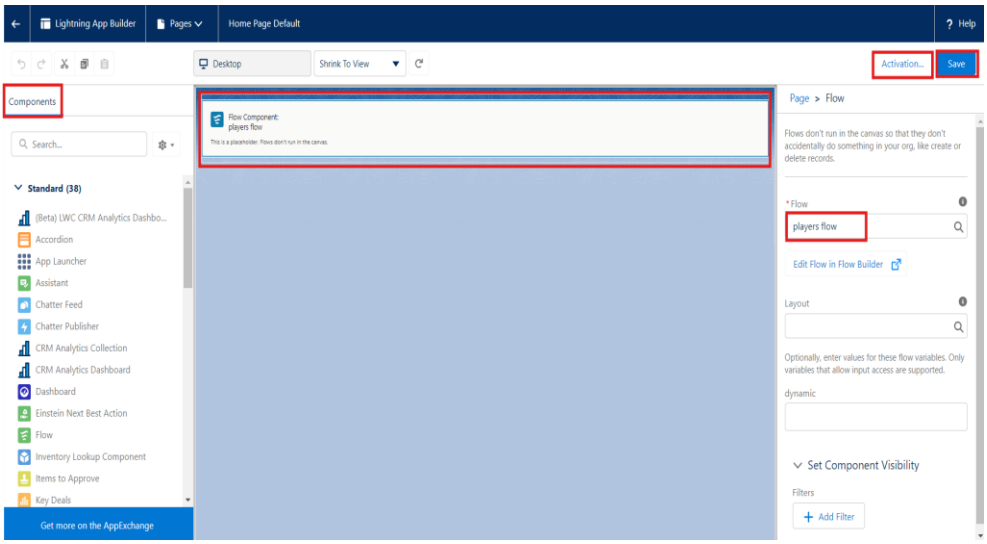


Activity 2: Flow on Home Page

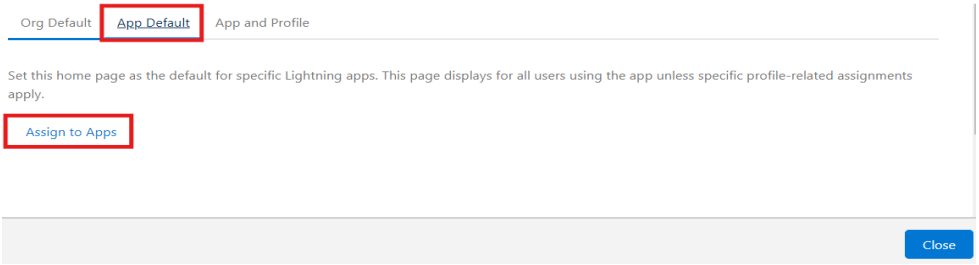
1. On your CRICKSTATS Application on the top right of the Navigation Bar click on **Personalize your nav bar icon (Pencil Icon)** and Add Home Page and click Save.



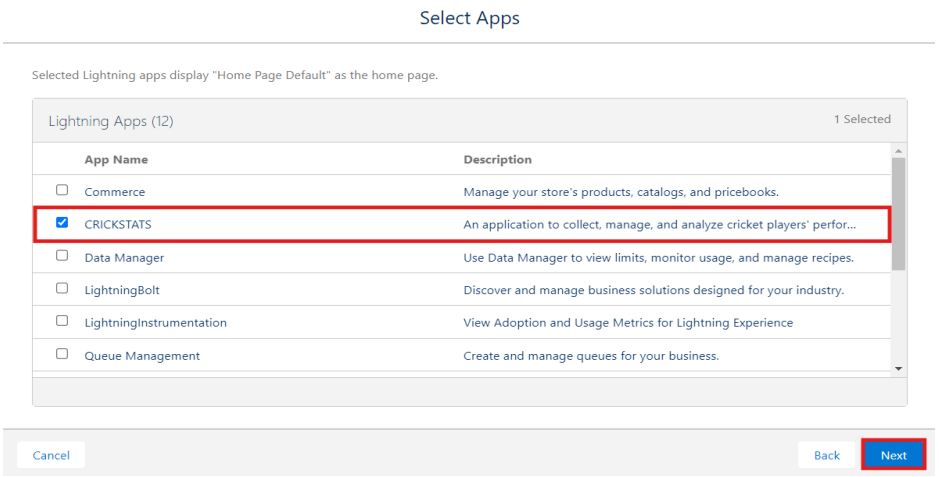
2. Remove all other components and Add **FLOW** component from the standard components on the left side.
3. **Flow** : Players Flow



4. Click **Save** and **Activation**.
5. Click on **App Default** and select **Assign to Apps.**



6. Select **CRICKSTAS** App and Click **Next** and then click **Save**.



7. FUNCTIONAL & PERFORMANCE TESTING

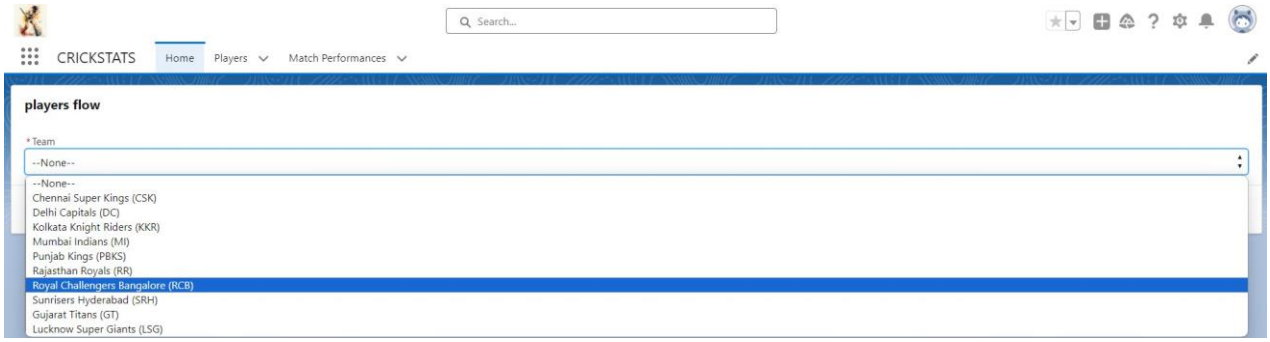
During the testing phase, screenshots were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system. These include record creation for Player and Match Performance objects; validation rule checks for mandatory fields, score limits, and input constraints; verification of record type behavior for Domestic and International match categories; and flow execution for automatic updating of player statistics and data consistency.

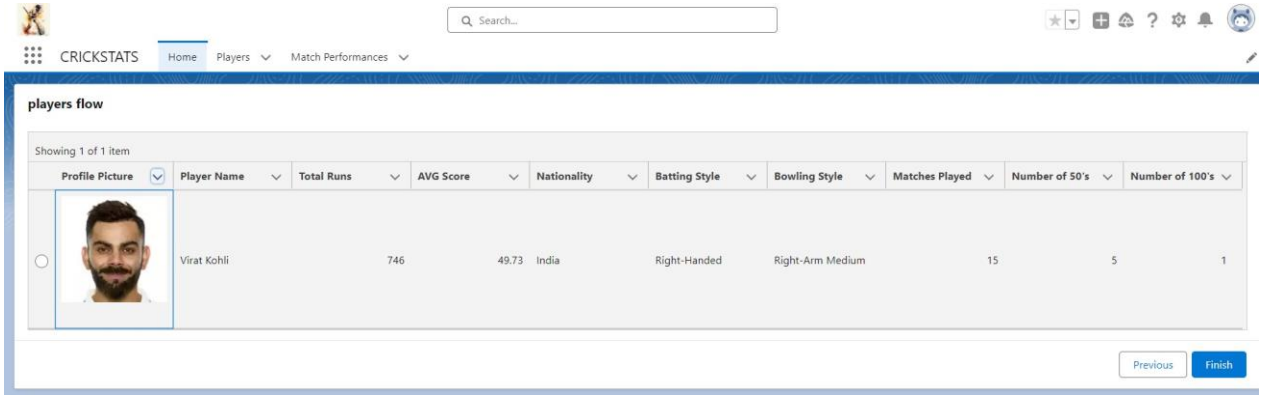
The functionality of roll-up summary fields was also tested to ensure accurate calculation of total runs, wickets, averages, and performance metrics. In addition, system performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading.

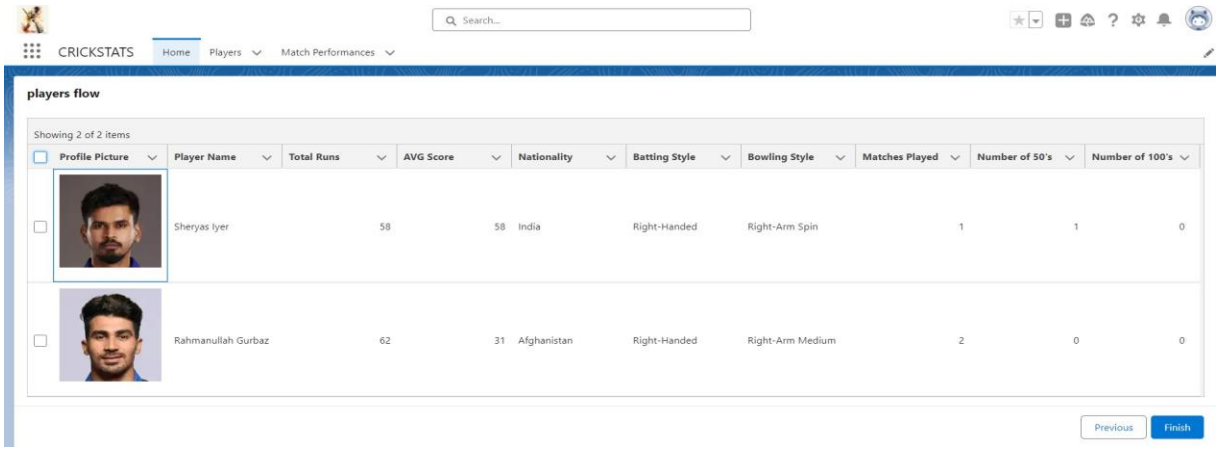
The application was tested under multiple user interactions to ensure stable performance without delays. Screenshots of generated reports and interactive dashboards were documented to confirm accurate data visualization, real-time updates, and reliable performance tracking. These testing activities ensured that the CRICKSTATS platform met both functional and performance

<u>NOTE</u>: Enter some records in the Player Object and Match Performance Object to get the results.

1. Select one Team from the Picklist and Click **Next**.







8. ADVANTAGES & DISADVANTAGES

Advantages

- The CRICKSTATS system provides several advantages compared to traditional manual record management systems.
- It offers a centralized platform where all player performance data is stored in a single location.
- The system provides real-time analytics, allowing coaches and analysts to evaluate player performance instantly.
- Security features ensure that sensitive data is protected from unauthorized access.
- The system is scalable, meaning it can support multiple teams, players, and tournaments. Automation reduces manual work and improves overall efficiency.
- Centralized system
- Real-time analytics
- Secure platform
- Scalable design
- Reduced manual work

Disadvantages

- Despite its advantages, the CRICKSTATS system also has some limitations.
- The application requires a Salesforce license, which may involve additional cost.
- The system depends on an internet connection since it is cloud-based.
- Offline access to the system is limited.
- Requires Salesforce license
- Internet dependency
- Limited offline access

9. Conclusion:

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making. The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development. The CRICKSTATS project successfully demonstrates how Salesforce technology can be used to develop a powerful cricket player performance tracking system. By digitizing player statistics and automating data management,

the system improves accuracy, efficiency, and accessibility of cricket performance records. The application also provides valuable insights through reports and dashboards, enabling coaches and team managers to make informed decisions about team strategy and player development.

10. FUTURE SCOPE

In the future, the CRICKSTATS system can be enhanced with additional features such as:

- Integration with mobile applications for real-time data entry
- Artificial intelligence models for predicting player performance
- Integration with video analysis tools for match review
- Development of a player ranking system
- Creation of a public portal where fans can view player statistics
- Support for multiple leagues and tournaments

These improvements will further enhance the capabilities of the system.

11. APPENDIX

- Salesforce Objects and Fields used in the project
- Player object details and field configurations
- Match Performance object structure
- Roll-up summary field configurations
- Salesforce Flow configurations used for automation
- Sample player and match performance records used for testing
- Reports and dashboard configurations for performance analysis
- Screenshots of system implementation and testing.

THANK YOU